THE HOMOTOPY THEORY OF MONOIDAL MODEL CATEGORIES

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Abstract. We prove that the ∞-category of presentably symmetric monoidal ∞-categories is equivalent to that of combinatorial symmetric monoidal model categories and symmetric monoidal left Quillen functors. This proves the conjecture by Pavlov. We also prove an analogous statement for non-symmetric monoidal semi-model categories, giving a partial solution to Hovey's 10th problem.

Size issues are pervasive in category theory, and careless handling of them can lead to pathological situations. For example, the theory of small categories with small limits is not categorically interesting, as they are merely preordered sets. However, a careful bookkeeping of size has a substantial payoff: It gives rise to the theory of *locally presentable categories*. These categories contain most large categories of interest, while enjoying good categorical properties (bicomplete, adjoint functor theorems hold, stable under exponential, etc.).

Presentable ∞-categories are an ∞-categorical generalization of locally presentable categories. As in the 1-categorical setting, they include most large ∞-categories of interest (spaces, spectra, derived ∞-categories, sheaves, etc.). At the same time, they retain excellent formal properties. For these reasons, they are now understood as a natural setting in which most homotopy-invariant constructions take place.

Historically, however, the presentable ∞-categories did not arise directly as a response to size issues in homotopy theory. Instead, they were predated by *combinatorial model categories*, which are model categories satisfying some size conditions. The advantage of combinatorial model categories is that many homotopy-theoretic constructions can be performed within an *ordinary* category, where many arguments and constructions are significantly more tractable than in ∞-categories. Because of this, they also remain a popular setting for homotopy-theoretic arguments.

This parallel development raises the question of how presentable ∞-categories and combinatorial model categories are related. In this direction, Dugger and Lurie showed that an ∞-category is presentable if and only if it underlies a combinatorial model category [Dug01a, Lur09]. More recently, Pavlov [Pav25] upgraded this result by showing that the homotopy theory of combinatorial model categories is equivalent to that of presentable ∞-categories. Informally, this establishes that combinatorial model categories provide concrete models for presentable ∞-categories.

In many cases, both ∞-categories and model categories come with symmetric monoidal structures. This naturally leads to the notions of *presentably symmetric monoidal ∞-categories* and *combinatorial symmetric monoidal model categories*. Based on his theorem, Pavlov conjectured that the homotopy theories of these two are likewise equivalent [Pav25, Conjecture 1.9].

On the one hand, establishing such an equivalence is important not only for conceptual completeness, but also because symmetric monoidal structures play a central role in modern homotopy theory, including operad theory, enrichment,

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and multiplicative constructions in general [WY18, Whi17, Whi22, WY24, BW22, BLW23, PS18a, PS18b, GH15, Hau15, Hei23, Hin20, Hin20, Lur17].

On the other hand, proving this conjecture is quite difficult, because very little is known about the homotopy theory of symmetric or non-symmetric monoidal model categories. For example, merely proving that every presentably symmetric monoidal ∞-categories is presented by a symmetric monoidal model category (i.e., a symmetric monoidal version of Dugger–Lurie's theorem) is nontrivial and had to rely on an ingenious method developed by Nikolaus–Sagave [NS17]. Even after their work, it wasn't known until the author's work [Ara26] that this holds monoidally. As another example, we do not even know whether every monoidal model category is equivalent to a simplicial one, which is the content of Hovery's 10th problem [Hov, Problem 10].

In this paper, we give a definitive solution to Pavlov's conjecture. Our main result is the following theorem:

Theorem A (Theorem 2.4)**.** *The functor*

$$L\colon \mathsf{CombSMMC}[\mathrm{Quillen.eq}^{-1}] \to \mathcal{P}\mathrm{r}\mathcal{SM}$$

is an equivalence, where:

- $\mathsf{CombSMMC}$ *denotes the category of combinatorial symmetric monoidal model categories and symmetric monoidal left Quillen functors;*
- Quillen.eq *denotes the class of Quillen equivalences;*
- $\mathcal{P}\mathrm{r}\mathcal{SM}$ *denotes the ∞-category of presentably symmetric monoidal ∞-categories; and*
- *L carries combinatorial symmetric monoidal model categories to their underlying symmetric monoidal ∞-category.*

Beyond its intrinsic theoretical importance, Theorem A provides an effective way to work with presentably symmetric monoidal ∞-categories. Roughly speaking, it ensures that if we want to prove a general statement about presentably symmetric monoidal ∞-categories, then we only need to prove the corresponding statements about combinatorial symmetric monoidal model categories. This reduction is very useful, because symmetric monoidal model categories are merely ordinary categories with additional structures, and various constructions are significantly more tractable there. In forthcoming work [Ara], we apply this strategy to establish a new equivalence of two models of enriched ∞-operads. The comparison uses Theorem A in a crucial way, and we believe that the comparison would be very difficult without it. We expect that there will be plenty of other applications of this sort.

Our approach to Theorem A is similar in spirit to that of Pavlov, but extending his argument to the symmetric monoidal setting presents new difficulties. To be precise, recall that his approach is built upon two inputs. The first is the work of Barwick–Kan [BK12b], which says that ∞-categories can be modeled by relative ∞-categories. The second is the work of Dugger and Low on presenting combinatorial model categories [Dug01a, Low16]. To prove Theorem A, we must extend both of these works to the symmetric monoidal setting. We will get over these problems by using author's earlier work on symmetric monoidal relative categories [Ara26] and the new model category of symmetric cubical sets (Subsection 5.1).

Remark 0.1. Theorem 2.4 admits many variations. Precise statements will appear in Section 6, but here are some of them stated informally:

- Every tractable (non-symmetric) monoidal semi-model category is monoidally Quillen equivalent to a simplicial one. This gives a partial solution to Hovey's 10th problem [Hov, Problem 10] (Remark 6.5).

- Every presentably symmetric monoidal ∞-category $\mathcal{C}$ admits an essentially unique presentation by a combinatorial symmetric monoidal model category (Remark 6.3).

Organization of the paper. This paper has 6 sections in the main body and 4 sections in the appendix.

- In Section 1, we recall the definition of monoidal model categories and their variations.
- In Section 2, we state the main theorem, and then show that it follows from three separate propositions. The proof of these propositions will be given in the next three sections (Section 3, 4, and 5).
- In Section 6, we give variations of the main result of this paper.
- Appendices contain miscellaneous results that will be used in the main body of the paper. In Appendix A, we give an overview of semi-model categories. In Appendix B, we review Gabriel–Ulmer duality and extend it multiplicatively. Appendix C discusses the compatibility of localization and base change. In Appendix D, we explain how to replace enriched monoidal model categories by ones with cofibrant unit.

Notation and convention.

- In addition to the ZFC axioms, we will assume the existence of three nested Grothendieck universes whose elements are called small sets, large sets, and very large sets.[1] All locally presentable categories are assumed to be large but not very large, so that the collection of locally presentable categories themselves form a very large set.
- We assume that model categories are at most large. Also, we only require the existence of finite limits and finite colimits for model categories. If $\mathbf{M}$ is a model category, we write $\mathbf{M}_{\mathrm{cof}} \subset \mathbf{M}$ for the full subcategory of cofibrant objects. We also write $\mathbf{M}_\infty$ for the underlying ∞-category of $\mathbf{M}$ (i.e., localization at weak equivalences).
- We use the term "regular cardinal" to mean "small regular cardinal."
- By ∞-categories, we mean *quasicategories* in the sense of [Joy02, Lur09]. We mostly follow the terminology and notation of [Lur09].
- If $\mathcal{C}$ is a presentable ∞-category and κ is a regular cardinal, we write $\mathcal{C}_\kappa \subset \mathcal{C}$ for the full subcategory spanned by the κ-compact objects of $\mathcal{C}$.
- We let Fin_* denote the category of finite pointed sets $\langle n \rangle = (\{*, 1, \ldots, n\}, *)$ for $n \geq 0$ and pointed maps.
- We let $\mathcal{SM}\mathsf{Cat}_\infty$ denote the ∞-category of small symmetric monoidal ∞-categories (defined as the underlying ∞-category of the model category described in [Lur17, Variant 2.1.4.13]). We write $\widehat{\mathcal{SM}\mathsf{Cat}}_\infty$ for the ∞-category of large symmetric monoidal ∞-categories.
- We write $\mathcal{PrSM} \subset \widehat{\mathcal{SM}\mathsf{Cat}}_\infty$ for the subcategory spanned by the presentably symmetric monoidal ∞-categories and symmetric monoidal functors that preserve small colimits. We define the ∞-category $\mathcal{Pr}\mathsf{Mon}$ of presentably monoidal ∞-categories similarly.
- We will not notationally distinguish between ordinary categories and their nerves. We will also regard every $(2,1)$-category (i.e., categories enriched over groupoids) as an ∞-category by taking its Duskin nerve. (Recall that this converts $(2,1)$-categories into ∞-categories, and it is functorial in strictly unitary pseudofunctors [Lur25, Tag 00AU].)

[1] Although Pavlov did not make this assumption in his paper [Pav25], it seems extremely inconvenient to drop this assumption due to the nature of the statements we will prove. For example, the statement of the main theorem itself needs to be adjusted without this.

- Given a pair of symmetric ∞-monoidal categories $\mathcal{C}, \mathcal{D}$, we let $\mathrm{Fun}^{\otimes}(\mathcal{C}, \mathcal{D})$ denote the ∞-category of symmetric monoidal functors $\mathcal{C} \to \mathcal{D}$. We use similar notation for symmetric monoidal categories.

1. Monoidal model categories and their variations

The use of the term "monoidal model category" and its variations is not entirely standardized in the literature. The goal of this section is to record the precise definitions we will use.

1.1. **Plain case.** We start with the definition of monoidal model categories. There are several competing definitions in the literature: Most definitions require the pushout-product axiom, but they often differ in their requirement on how "flat" the unit object should be. In some literature (e.g., [Lur17]), the unit object is required to be cofibrant. This assumption is convenient for theoretical purposes but ends up excluding many interesting examples, such as symmetric spectra. At the other extreme (e.g., [PS18a]), no condition on the unit object is assumed. In this paper, we go for a middle ground by adopting Muro's unit axiom, which is satisfied by most monoidal model categories we encounter. (The author is not aware of a counterexample.)

Definition 1.1. Let $\mathbf{A}, \mathbf{B}, \mathbf{C}$ be model categories. A functor $F\colon \mathbf{A} \times \mathbf{B} \to \mathbf{C}$ is called a **left Quillen bifunctor** if it satisfies the following pair of conditions:

(1) The functor F preserves small colimits in each variable.
(2) For every pair of cofibrations $f\colon A \to A'$ in $\mathbf{A}$ and $g\colon B \to B'$ in $\mathbf{B}$, the map
$$F(A, B') \amalg_{F(A,B)} F(A', B) \to F(A, B)$$
is a cofibration, and moreover it is a weak equivalence if f or g is one.

Definition 1.2.

- A **monoidal model category** is a biclosed monoidal category $(\mathbf{M}, \otimes, \mathbf{1})$ equipped with a model structure, satisfying the following axioms:
  (1) (**Pushout-product axiom**) The functor $\otimes\colon \mathbf{M} \times \mathbf{M} \to \mathbf{M}$ is a left Quillen bifunctor (Definition 1.1).
  (2) (**Muro's unit axiom** [Mur15]) There is a weak equivalence $q\colon \widetilde{\mathbf{1}} \to \mathbf{1}$, with $\widetilde{\mathbf{1}}$ cofibrant, such that for every object $X \in \mathbf{M}$, the maps $X \otimes q$ and $q \otimes X$ are weak equivalences.
- We say that a monoidal model category is **combinatorial** if its underlying model category is combinatorial. If further it admits generating sets of cofibrations and trivial cofibrations whose elements have a cofibrant domain, it is called **tractable**.
- If $\mathbf{M}$ and $\mathbf{N}$ are monoidal model categories, then a **monoidal left Quillen functor** is a monoidal functor[2] $\mathbf{M} \to \mathbf{N}$ whose underlying functor of model categories is left Quillen. We will write $\mathrm{Fun}^{\otimes, LQ}(\mathbf{M}, \mathbf{N})$ for the category of monoidal left Quillen functors $\mathbf{M} \to \mathbf{N}$ and monoidal natural transformations between them.
- We will write $\mathsf{CombMMC}$ for the (very large) category of combinatorial monoidal model categories and monoidal left Quillen functors, and write $\mathsf{CombMMC}^{\mathbf{1}} \subset \mathsf{CombMMC}$ for the full subcategory spanned by the combinatorial monoidal model categories with a cofibrant unit.

[2] By a monoidal functor, we mean a strong monoidal functor in the sense of [ML98, Chapter XI].

- **Symmetric monoidal model categories** and **symmetric monoidal left Quillen functors** are defined similarly. (We will use the notation $\mathrm{Fun}^{\otimes,LQ}(-,-)$ in the symmetric monoidal case, too. This is abusive but is rarely confusing.) The categories $\mathsf{CombSMMC}$ and $\mathsf{CombSMMC}^{\mathbf{1}}$ are also defined similarly, using combinatorial symmetric monoidal model categories.
- We define categories $\mathsf{TractMMC}$, $\mathsf{TractSMMC}^{\mathbf{1}}$, etc, similarly, using tractable symmetric and non-symmetric monoidal model categories.

The reason why we adopt Muro's unit axiom is that it allows us to turn every monoidal model category into one with a cofibrant unit. More precisely, we have the following theorem.

Theorem 1.3. [Mur15, Theorem 1, Proposition 12] *The inclusion* $\mathsf{CombMMC}^{\mathbf{1}} \subset \mathsf{CombMMC}$ *admits a left adjoint. Moreover, the unit of this adjunction preserves and reflects weak equivalences and induces an isomorphism of underlying monoidal categories. A similar claim holds for combinatorial symmetric monoidal model categories.*

The categories $\mathsf{CombMMC}$ can be upgraded to a $(2,1)$-category $\mathsf{CombMMC}_{(2,1)}$, whose mapping groupoid are given by the maximal subgroupoid $\mathrm{Fun}^{\otimes,LQ}(\mathbf{M},\mathbf{N})^{\cong} \subset \mathrm{Fun}^{\otimes,LQ}(\mathbf{M},\mathbf{N})$. We will use similar notations for the categories introduced in Definition 1.2. These ∞-categories satisfy the following universal property:

Proposition 1.4. *The functors*

$$\begin{aligned}\mathsf{CombMMC} &\to \mathsf{CombMMC}_{(2,1)},\\ \mathsf{CombMMC}^{\mathbf{1}} &\to \mathsf{CombMMC}^{\mathbf{1}}_{(2,1)},\\ \mathsf{CombSMMC} &\to \mathsf{CombSMMC}_{(2,1)},\\ \mathsf{CombSMMC}^{\mathbf{1}} &\to \mathsf{CombSMMC}^{\mathbf{1}}_{(2,1)}\end{aligned}$$

are ∞-categorical localizations at the left Quillen functors whose underlying functors are equivalences of categories.

Proof. We will prove the assertion for the first functor; the remaining assertions can be proved in a similar way. According to (the dual of) [ACK25, Corollary 3.16], it suffices to show that $\mathsf{CombMMC}_{(2,1)}$ admits weak cotensor by [1]. That is, it suffices to show that for each $\mathbf{N} \in \mathsf{CombMMC}_{(2,1)}$, there is an object $[1] \pitchfork \mathbf{N}$ admitting an isomorphism of groupoids

$$\mathrm{Fun}^{\otimes,LQ}(\mathbf{M},[1] \pitchfork \mathbf{N})^{\cong} \cong \mathrm{Fun}\Big([1], \mathrm{Fun}^{\otimes,LQ}(\mathbf{M},\mathbf{N})^{\cong}\Big)$$

natural in $\mathbf{M} \in \mathsf{CombMMC}$. For this, we consider the groupoid J with two objects $0,1$, and with exactly one morphism between any pair of objects. We can make $\mathbf{N}^J$ into a monoidal model category by using the degreewise tensor product and the model structure coming from the equivalence of categories $\mathbf{N} \xrightarrow{\simeq} \mathbf{N}^J$. Then $\mathbf{N}^J$ is a weak cotensor of $\mathbf{N}$ by [1], and we are done. □

1.2. **Enriched case.** We next turn to the definition of enriched monoidal model categories.

Definition 1.5. Let $\mathbf{V}$ be a combinatorial symmetric monoidal model category.

- A **$\mathbf{V}$-monoidal model category** is a tensored and cotensored $\mathbf{V}$-monoidal category $\mathbf{M}$ (in the sense of [Day70]) which is biclosed (i.e., for each $X \in \mathbf{M}$, the $\mathbf{V}$-functors $X \otimes -$ and $- \otimes X$ admit right adjoints) and equipped with a model structure on its underlying category, such that the tensor bifunctor

$$\mathbf{V} \times \mathbf{M} \to \mathbf{M}$$

is a left Quillen bifunctor.

- We say that a $\mathbf{V}$-monoidal model category is **combinatorial** if its underlying model category is combinatorial.
- If $\mathbf{M}$ and $\mathbf{N}$ are $\mathbf{V}$-monoidal model categories, we write $\mathrm{Fun}^{\otimes,LQ}_{\mathbf{V}}(\mathbf{M},\mathbf{N})$ for the category of $\mathbf{V}$-monoidal functors $\mathbf{M}\to\mathbf{N}$ that are $\mathbf{V}$-cocontinuous and whose underlying functors are left Quillen.
- We write $\mathsf{CombMMC}_{\mathbf{V},(2,1)}$ for the $(2,1)$-category whose objects are the combinatorial $\mathbf{V}$-monoidal model categories, with hom-groupoids given by the maximal subgroupoids of $\mathrm{Fun}^{\otimes,LQ}_{\mathbf{V}}(-,-)$.
- We write $\mathsf{CombMMC}_{\mathbf{V}}$ for the underlying category of $\mathsf{CombMMC}_{\mathbf{V},(2,1)}$.

We define $\mathbf{V}$-symmetric monoidal model categories similarly, and define a $(2,1)$-category $\mathsf{CombSMMC}_{\mathbf{V},(2,1)}$ and an ordinary category $\mathsf{CombSMMC}_{\mathbf{V}}$ similarly. As in Definition 1.2, we also consider full subcategories of these categories such as $\mathsf{TractMMC}_{\mathbf{V}}$ and $\mathsf{TractSMMC}^{\mathbf{1}}_{\mathbf{V}}$.

In ordinary algebra, a commutative algebra A over a commutative ring k can be defined as a commutative ring equipped with a ring homomorphism $k\to A$. Extending an analogy of this to the model-categorical setting, we arrive at the notion of *symmetric algebras*:

Definition 1.6. Let $\mathbf{V}$ be a combinatorial symmetric monoidal model category.

- A **symmetric $\mathbf{V}$-algebra** is a monoidal model category $\mathbf{M}$ equipped with a monoidal left Quillen functor $\mathbf{V}\to\mathbf{M}$.
- We say that a symmetric $\mathbf{V}$-algebra is **combinatorial** if its underlying model category is combinatorial.
- We write $\mathsf{CombSym}\mathbf{V}\text{-}\mathsf{Alg}_{(2,1)} = \int^{\mathbf{M}\in\mathsf{CombSMMC}_{(2,1)}} \mathrm{Fun}^{\otimes,LQ}(\mathbf{V},\mathbf{M})$ for the $(2,1)$-category of combinatorial symmetric $\mathbf{V}$-algebras, where $\int$ denotes the 2-categorical Grothendieck construction [Str80, 1.10]. Explicitly:
  - Objects are combinatorial symmetric $\mathbf{V}$-algebras.
  - A 1-morphism $(F\colon\mathbf{V}\to\mathbf{M})\to(G\colon\mathbf{V}\to\mathbf{N})$ is a pair (H,α), where $H\colon\mathbf{M}\to\mathbf{N}$ is a symmetric monoidal functor and $\alpha\colon HF\Rightarrow G$ is a symmetric monoidal natural transformation.
  - A 2-morphism $(H,\alpha)\Rightarrow(H',\alpha')\colon(F\colon\mathbf{V}\to\mathbf{M})\to(G\colon\mathbf{V}\to\mathbf{N})$ is a symmetric monoidal natural isomorphism $\beta\colon H\overset{\cong}{\Rightarrow}H'$ satisfying $\alpha'\circ\beta F=\alpha$:

$$\begin{tikzcd}
 & \mathbf{V} \arrow[dl, "F"'] \arrow[dr, "G"] \arrow[d, phantom, "=\alpha\Rightarrow"] & \\
\mathbf{M} \arrow[rr, "H"'] & {} & \mathbf{N}
\end{tikzcd}
\quad = \quad
\begin{tikzcd}
 & \mathbf{V} \arrow[dl, "F"'] \arrow[dr, "G"] \arrow[d, phantom, "=\alpha'\Rightarrow"] & \\
\mathbf{M} \arrow[rr, bend left, "H'"] \arrow[rr, bend right, "H"'] & {} \arrow[d, phantom, "\overset{\wedge}{\beta}\,\shortparallel"] & \mathbf{N}.
\end{tikzcd}$$

- We write $\mathsf{CombSym}\mathbf{V}\text{-}\mathsf{Alg}$ for the underlying category of $\mathsf{CombSym}\mathbf{V}\text{-}\mathsf{Alg}_{(2,1)}$.

Remark 1.7. There is a related notion of *central algebras* [Hov99, Definition 4.1.10], but they are not amenable to the techniques we use in this paper.

As in Proposition 1.4, we have:

Proposition 1.8. *Let* $\mathbf{V}$ *be a combinatorial symmetric monoidal model category. The functors*

$$\begin{aligned}
\mathsf{CombMMC}_{\mathbf{V}} &\to \mathsf{CombMMC}_{\mathbf{V},(2,1)},\\
\mathsf{CombMMC}^{\mathbf{1}}_{\mathbf{V}} &\to \mathsf{CombMMC}^{\mathbf{1}}_{\mathbf{V},(2,1)},\\
\mathsf{CombSMMC}_{\mathbf{V}} &\to \mathsf{CombSMMC}_{\mathbf{V},(2,1)},\\
\mathsf{CombSMMC}^{\mathbf{1}}_{\mathbf{V}} &\to \mathsf{CombSMMC}^{\mathbf{1}}_{\mathbf{V},(2,1)},\\
\mathsf{CombSym}\mathbf{V}\text{-}\mathsf{Alg} &\to \mathsf{CombSym}\mathbf{V}\text{-}\mathsf{Alg}_{(2,1)}
\end{aligned}$$

are all localizations.

1.3. **Semi-model-categorical case.** (Left) **Semi-model categories** are a weakening of model categories in which part of the axioms of lifting and factorization are only required to hold for cofibrations with *cofibrant* source [BW24, Definition 2.1]. While this leads to a less attractive set of axioms, semi-model categories are practically indistinguishable from model categories. The main advantages of semi-model categories over ordinary model categories is that they are much easier to construct than a full model structure. In particular, tractable semi-model categories are fully compatible with (left) Bousfield localization [BW24]. For these reasons, semi-model categories is becoming increasingly popular, and we have no reason not to include them in this paper.

For the basics of semi-model categories, we refer the reader to Section A. In this subsection, we collect definitions related to the interplay of semi-model categories and monoidal structures.

Definition 1.9. A semi-model category is said to be **tractable** if it is locally presentable as a category and admits generating sets of cofibrations of cofibrant objects and trivial cofibrations of cofibrant objects.

Definition 1.10. Let $\mathbf{M}, \mathbf{N}, \mathbf{P}$ be tractable semi-model categories. A **left Quillen bifunctor** $F\colon \mathbf{M}\times\mathbf{N} \to \mathbf{P}$ is a functor that preserves small colimits in each variable, and which satisfies the pushout-product axiom for cofibrations of cofibrant objects and trivial cofibrations of cofibrant objects.

Definition 1.11. The definition of monoidal model categories carries over to the semi-model categorical case. We define a category $\mathsf{TractMMC}_{\mathrm{semi}}$ of tractable monoidal semi-model categories, and define categories such as $\mathsf{TractMMC}^{\mathbf{1}}_{\mathrm{semi}}$, $\mathsf{TractSMMC}_{\mathrm{semi}}$, and $\mathsf{TractSMMC}^{\mathbf{1}}_{\mathrm{semi}}$ similarly, as in Definition 1.2. Also, for a tractable symmetric monoidal semi-model category $\mathbf{V}$, there is a notion of $\mathbf{V}$-monoidal model categories and $\mathbf{V}$-symmetric monoidal model categories. The categories $\mathsf{TractMMC}^{\mathbf{1}}_{\mathbf{V},\mathrm{semi}}$ and $\mathsf{TractSMMC}_{\mathbf{V},\mathrm{semi}}$, etc, are defined exactly as in Definition 1.5.

Remark 1.12. Theorem 1.3 and Proposition 1.4 remains valid for semi-model categories, with the same proof.

2. Main result

In this section, we state the main result of this paper and explain how we will prove it.

To state our main result, we need a bit of terminology and notation.

Definition 2.1. [Ara26, Definition 0.1] A **symmetric monoidal relative category** is a symmetric monoidal category $\mathbf{C}$ equipped with a subcategory $\mathbf{W} \subset \mathbf{C}$ that contains all isomorphisms and is stable under tensor products. Morphisms of $\mathbf{W}$ are called **weak equivalences**.

We write $\mathsf{SMRelCat}$ for the category of small symmetric monoidal relative categories and symmetric monoidal relative functors that preserve weak equivalences.

We define the category $\widehat{\mathsf{SMRelCat}}$ of *large* symmetric monoidal relative categories similarly.

Example 2.2. Let $\mathbf{M}$ be a symmetric monoidal model category. Weak equivalences of $\mathbf{M}$ are generally not stable under tensor products, so $\mathbf{M}$ is generally *not* a symmetric monoidal relative category. Nonetheless, its full subcategory $\mathbf{M}^\flat \subset \mathbf{M}$ of cofibrant objects and the objects isomorphic to the unit object is a symmetric monoidal relative category.

For the next definition, we recall that localization of symmetric monoidal relative categories determines a functor $L\colon \mathsf{SMRelCat} \to \mathcal{SMC}\mathsf{at}_\infty$ [Ara26, Notation 2.1].

Definition 2.3. We define a functor

$$(-)_\infty \colon \mathsf{CombSMMC} \to \mathcal{PrSM}$$

as the codomain restriction of the composite

$$\mathsf{CombSMMC} \xrightarrow{(-)^\flat} \widehat{\mathsf{SMRelCat}} \xrightarrow{L} \widehat{\mathcal{SMC}\mathsf{at}}_\infty.$$

(Note that this is well-defined by [Lur17, Proposition 1.3.4.22 and Corollary 1.3.4.26].) If $\mathbf{M}$ is a combinatorial symmetric monoidal model category, we refer to $\mathbf{M}_\infty$ as its **underlying symmetric monoidal ∞-category**.

We now arrive at the statement of the main result.

Theorem 2.4. *The functor $(-)_\infty$ of Definition 2.3 is a localization at Quillen equivalences, i.e., it induces a categorical equivalence*

$$\mathsf{CombSMMC}[\mathrm{Quill.eq}^{-1}] \xrightarrow{\simeq} \mathcal{PrSM}.$$

We will prove Theorem 2.4 as follows: Fix a left proper tractable symmetric monoidal model category $\mathbf{S}$ with a cofibrant unit, whose underlying symmetric monoidal ∞-category is equivalent to the cartesian monoidal ∞-category $\mathcal{S}$ of ∞-groupoids.[3] (One example is the cartesian model category of simplicial sets with the Kan–Quillen model structure.) We also define a category $\widehat{\mathsf{SMRelCat}}_{\mathcal{PrSM}}$ by the pullback

$$\begin{tikzcd} \widehat{\mathsf{SMRelCat}}_{\mathcal{PrSM}} \arrow[r] \arrow[d] \arrow[dr, phantom, "\lrcorner", very near start] & \widehat{\mathsf{SMRelCat}} \arrow[d, "L"] \\ \mathcal{PrSM} \arrow[r] & \widehat{\mathcal{SMC}\mathsf{at}}_\infty. \end{tikzcd}$$

We then contemplate the following commutative diagram

$$\begin{tikzcd} \mathsf{CombSym}\mathbf{S}\text{-}\mathsf{Alg} \arrow[r, "U"] \arrow[d, "(-)_\flat"'] & \mathsf{CombSMMC}^{\mathbf{1}} \arrow[r, hook, "\iota"] & \mathsf{CombSMMC} \arrow[d, "(-)_\infty"] \\ \widehat{\mathsf{SMRelCat}}_{\mathcal{PrSM}} \arrow[rr, "L"'] & & \mathcal{PrSM}\widehat{\mathcal{C}\mathsf{at}}_\infty, \end{tikzcd}$$

where U is the forgetful functor and ι is the inclusion. We will then prove the following propositions:

Proposition 2.5. *The functor $L\colon \widehat{\mathsf{SMRelCat}}_{\mathcal{PrSM}} \to \mathcal{PrSM}\widehat{\mathcal{C}\mathsf{at}}_\infty$ induces a categorical equivalence*

$$\widehat{\mathsf{SMRelCat}}_{\mathcal{Pr}}[\mathrm{loc.eq}^{-1}] \to \mathcal{PrSM}\widehat{\mathcal{C}\mathsf{at}}_\infty,$$

where loc.eq *denotes the subcategory of* ***local equivalences****, i.e., morphisms inverted by L.*

[3] Since $\mathcal{S}$ is the initial presentably symmetric monoidal ∞-category [Lur17, Corollary 3.2.1.9, Example 4.8.1.20], such an equivalence is unique up to a contractible space of choices if it exists.

Proposition 2.6. *The functor* $(-)_\flat$ *induces a categorical equivalence*

$$\mathsf{CombSym}\mathbf{S}\text{-}\mathsf{Alg}[\mathrm{Quill.eq}^{-1}] \xrightarrow{\simeq} \mathsf{SMRel}\widehat{\mathsf{Cat}}_{\mathcal{P}r\mathcal{SM}}[\mathrm{loc.eq}^{-1}].$$

Proposition 2.7. *The functor* U *induces a categorical equivalence*

$$\mathsf{CombSym}\mathbf{S}\text{-}\mathsf{Alg}[\mathrm{Quill.eq}^{-1}] \xrightarrow{\simeq} \mathsf{CombSMMC}^{\mathbf{1}}[\mathrm{Quill.eq}^{-1}].$$

The proof of these propositions will be given in the next three sections (Section 3, 4, and 5). Since we know that ι induces a categorical equivalence upon localizing at Quillen equivalences (Theorem 1.3), these proposition will prove Theorem 2.4.

3. Proof of Proposition 2.5

In this section, we prove Proposition 2.5, which asserts that the functor

$$\mathsf{SMRel}\widehat{\mathsf{Cat}}_{\mathcal{P}r\mathcal{SM}}[\mathrm{loc.eq}^{-1}] \to \mathcal{P}r\mathcal{SM}$$

is an equivalence. In fact, we will prove the following more general assertion:

Lemma 3.1. *For every conservative functor* $\mathcal{X} \to \mathcal{SMC}\mathsf{at}_\infty$ *of* ∞*-categories, the functor*

$$\mathsf{SMRelCat}_{\mathcal{X}} = \mathcal{X} \times_{\mathcal{SMC}\mathsf{at}_\infty} \mathsf{SMRelCat} \to \mathcal{X}$$

is a localization.

Replacing $\mathcal{SMC}\mathsf{at}_\infty$ by $\mathcal{SM}\widehat{\mathcal{C}\mathsf{at}}_\infty$ and then substituting $\mathcal{P}r\mathcal{SM}$ for $\mathcal{X}$, we obtain Proposition 2.5.

For the proof of Lemma 3.1, we need some terminology and notation.

- A **relative ∞-category** is an ∞-category equipped with a subcategory $\mathcal{W} \subset \mathcal{C}$ containing all equivalences. Morphisms in $\mathcal{W}$ are typically called **weak equivalences**. When $\mathcal{C}$ is (the nerve of) an ordinary category, we call such a pair a **relative category**. This should not be confused with the definition of Barwick–Kan [BK12b], which only requires that $\mathcal{W}$ contains all objects.
- If $\mathcal{C}$ and $\mathcal{D}$ are relative ∞-categories, a **relative functor** $f\colon \mathcal{C} \to \mathcal{D}$ is a functor between the underlying ∞-categories that preserve weak equivalences. We say that such an f is a **homotopy equivalence** if there is a relative functor $g\colon \mathcal{D} \to \mathcal{C}$ and zig-zags of natural weak equivalences connecting $f \circ g$ and $g \circ f$ to the identity functors.
- We write RelCat for the category of small relative categories and functors preserving weak equivalences. For disambiguation, we will write $\mathsf{RelCat}_{\mathrm{BK}}$ for the slightly larger category of Barwick–Kan's relative categories that are small.
- We call a morphsim $f\colon \mathcal{C} \to \mathcal{D}$ of $\mathsf{RelCat}_{\mathrm{BK}}$ a **local equivalence** if it induces an equivalence between the (∞-categorical) localizations.
- Recall that a symmetric monoidal category is called a **permutative category** if its underlying monoidal category is strict (i.e., its associators and the unitors are the identity maps). We write $\mathsf{PermRelCat} \subset \mathsf{SMRelCat}$ for the subcategory of objects whose underlying symmetric monoidal categories are permutative categories, and morphisms whose underlying symmetric monoidal functors are strict.
- We will regard the categories $\mathsf{RelCat}, \mathsf{RelCat}_{\mathrm{BK}}, \mathsf{PermRelCat}, \mathsf{SMRelCat}$ as relative categories whose weak equivalences are the local equivalences.
- We write Fin_* for the category of finite pointed sets and pointed maps. We write $\mathrm{Fun}^{\mathrm{Seg}}(\mathsf{Fin}_*, \mathsf{RelCat}) \subset \mathrm{Fun}^{\mathrm{Seg}}(\mathsf{Fin}_*, \mathsf{RelCat})$ for the full subcategory

spanned by the functors $F\colon \mathsf{Fin}_* \to \mathsf{RelCat}$ satisfying the following **Segal condition**: For every $n \geq 0$, the map $\rho_i\colon \langle n\rangle \to \langle 1\rangle$ defined by

$$\rho_i(j) = \begin{cases} * & \text{if } i \neq j, \\ 1 & \text{if } i = j, \end{cases}$$

induces a local equivalence

$$F\langle n\rangle \xrightarrow{\simeq} \prod_{1\leq i\leq n} F\langle 1\rangle.$$

We define $\mathrm{Fun}^{\mathrm{Seg}}(\mathsf{Fin}_*, \mathsf{RelCat}_{\mathrm{BK}}) \subset \mathrm{Fun}^{\mathrm{Seg}}(\mathsf{Fin}_*, \mathsf{RelCat}_{\mathrm{BK}})$ similarly.

Proof of Lemma 3.1. Since symmetric monoidal categories are functorially equivalent to permutative categories [May78, Proposition 4.2], using Proposition C.3, we may replace $\mathsf{SMRelCat}$ by $\mathsf{PermRelCat}$.

According to [Ara26, Theorem 1.1], there is a homotopy equivalence $\mathrm{Fact}\colon \mathsf{PermRelCat} \to \mathrm{Fun}^{\mathrm{Seg}}(\mathsf{Fin}_*, \mathsf{RelCat})$ of relative categories. The proof of loc. cit. further shows that the localization of $\mathrm{Fun}^{\mathrm{Seg}}(\mathsf{Fin}_*, \mathsf{RelCat})$ is equivalent to $\mathcal{SMC}\mathsf{at}_\infty$, and the diagram

$$\begin{tikzcd} \mathsf{PermRelCat} \arrow[r, "\mathrm{Fact}"] \arrow[d, hook] & \mathrm{Fun}^{\mathrm{Seg}}(\mathsf{Fin}_*, \mathsf{RelCat}) \arrow[d, "\text{localization}"] \\ \mathsf{SMRelCat} \arrow[r, "L"'] & \mathcal{SMC}\mathsf{at}_\infty. \end{tikzcd}$$

commutes up to natural equivalence. Thus, by Proposition C.3, it will suffice to show that the functor

$$\mathrm{Fun}^{\mathrm{Seg}}(\mathsf{Fin}_*, \mathsf{RelCat})_{\mathcal{X}} \to \mathcal{X}$$

is a localization. Since the inclusion $\mathsf{RelCat} \hookrightarrow \mathsf{RelCat}_{\mathrm{BK}}$ is a homotopy equivalence of relative categories, we can replace RelCat by $\mathsf{RelCat}_{\mathrm{BK}}$ by Proposition C.3 again.

We now recall from [BK12b, Theorem 6.1] and [BK12a, Theorem 1.8] that $\mathsf{RelCat}_{\mathrm{BK}}$ and its local equivalences is part of a combinatorial model structure. Therefore, Theorem C.1 and [Hir03, Theorem 11.6.1] show that the localization

$$\mathrm{Fun}(\mathsf{Fin}_*, \mathsf{RelCat}_{\mathrm{BK}}) \to \mathrm{Fun}(\mathsf{Fin}_*, \mathsf{RelCat}_{\mathrm{BK}})[\mathrm{weq}^{-1}]$$

is stable under pullback. In particular, the functor

$$\mathrm{Fun}^{\mathrm{Seg}}(\mathsf{Fin}_*, \mathsf{RelCat}_{\mathrm{BK}})_{\mathcal{X}} \to \mathcal{X}$$

is a localization, which was to be proved. □

4. Proof of Proposition 2.6

Notation 4.1. Throughout this section, we fix a left proper tractable symmetric monoidal model category $\mathbf{S}$ with a cofibrant unit, whose underlying symmetric monoidal ∞-category is equivalent to the cartesian monoidal ∞-category $\mathcal{S}$ of ∞-groupoids.

In this section, we give a proof of Proposition 2.6, which asserts that the functor

$$\mathsf{CombSym}\mathbf{S}\text{-}\mathsf{Alg}[\mathrm{Quill.eq}^{-1}] \xrightarrow{\simeq} \widehat{\mathsf{SMRelCat}}_{\mathcal{P}\mathrm{r}\mathcal{SM}}[\mathrm{loc.eq}^{-1}]$$

is an equivalence. This will need two ingredients: A relative-categorical version of the multiplicative Gabriel–Ulmer duality, and some formal cardinality argument of model categories. We will tackle each of these in the next two subsections, and then prove Proposition 2.6 in the final subsection.

4.1. **Relative-categorical multiplicative Gabriel–Ulmer duality.** In this subsection, we state and prove a relative-categorical multiplicative Gabriel–Ulmer duality (Proposition 4.6). We refer the reader to Section B for a review of Gabriel–Ulmer duality.

To state the main result, we need a bit of preliminaries.

Notation 4.2. Let κ be an uncountable regular cardinal. We write $\mathcal{SMC}\mathsf{at}_\infty(\kappa)$ for the subcategory of $\mathcal{SMC}\mathsf{at}_\infty$ spanned by the symmetric monoidal ∞-categories that are compatible with κ-small colimits (i.e., those ∞-categories that have κ-small colimits and whose tensor product preserves such colimits in each variable).

Construction 4.3. Let κ be a regular cardinal. For each $\mathcal{C} \in \mathsf{SMRelCat}_{\mathcal{SMC}\mathsf{at}_\infty(\kappa)}$, we define an $\mathbf{S}$-algebra $\mathbf{Ind}^{\mathbf{S}}_\kappa(L(\mathcal{C}))$ as follows:

- Its underlying category is $\mathrm{Fun}(\mathcal{C}^{\mathrm{op}}, \mathbf{S})$.
- The symmetric monoidal structure comes from Day's convolution product.
- The symmetric monoidal functor $\mathbf{S} \to \mathrm{Fun}(\mathcal{C}^{\mathrm{op}}, \mathbf{S})$ is given by $X \mapsto \mathcal{C}(-, \mathbf{1}) \cdot X$, where $\mathbf{1} \in \mathcal{C}$ denote the monoidal unit and the dot "$\cdot$" denotes copowering by sets.
- The model structure is the left Bousfield localization of the projective model structure on $\mathrm{Fun}(\mathcal{C}^{\mathrm{op}}, \mathbf{S})$ whose weak equivalences are the maps inverted by the composite

$$\mathrm{Fun}(\mathcal{C}^{\mathrm{op}}, \mathbf{S}) \to \mathrm{Fun}(\mathcal{C}^{\mathrm{op}}, \mathcal{S}) \to \mathrm{Fun}(L(\mathcal{C})^{\mathrm{op}}, \mathcal{S}) \to \mathrm{Ind}_\kappa(L(\mathcal{C})).$$

  Here, the first functor postcomposes with the localization $\mathbf{S} \to \mathcal{S}$, the second functor is the left adjoint Kan extension along the localization functor $\mathcal{C}^{\mathrm{op}} \to L(\mathcal{C})^{\mathrm{op}}$, and the third functor is the left adjoint to the inclusion $\mathrm{Ind}_\kappa(L(\mathcal{C})) \hookrightarrow \mathrm{Fun}(L(\mathcal{C}), \mathcal{S})$.

We sometimes use the notation $\mathbf{Ind}^{\mathbf{S}}_\kappa(L(\mathcal{C}))$ for essentially small symmetric monoidal relative categories whose symmetric monoidal localization is idempotent complete and is compatible with κ-small colimits.

Lemma 4.4. *Construction 4.3 is well-defined. More precisely, in the situation of Construction 4.3, the following holds:*

(1) The model category $\mathbf{Ind}^{\mathbf{S}}_\kappa(L(\mathcal{C}))$ exists.
(2) The model category $\mathbf{Ind}^{\mathbf{S}}_\kappa(L(\mathcal{C}))$ satisfies the pushout-product axiom, and the monoidal unit is cofibrant.
(3) The symmetric monoidal functor $\mathbf{S} \to \mathbf{Ind}^{\mathbf{S}}_\kappa(L(\mathcal{C}))$ is left Quillen.

Proof. Part (1) is a consequence of Proposition A.20.

Next, we prove part (2). Choose generating sets I, J of cofibrations and trivial cofibrations of $\mathbf{S}$. According to [Hir03, Theorem 11.6.1], the sets $I_{\mathcal{C}} = \{\mathcal{C}(-, C) \cdot i\}_{i \in I, C \in \mathcal{C}}$ and $J_{\mathcal{C}} = \{\mathcal{C}(-, C) \cdot j\}_{j \in J, C \in \mathcal{C}}$ generate cofibrations and trivial cofibrations of the projective model structure on $\mathrm{Fun}(\mathcal{C}^{\mathrm{op}}, \mathbf{S})$. It follows immediately that:

- The monoidal unit of $\mathrm{Fun}(\mathcal{C}^{\mathrm{op}}, \mathbf{S})$ is cofibrant (because $\mathbf{S}$ has a cofibrant unit).
- The Day convolution product satisfies the pushout-product axiom for the projective model structure.
- The projective model structure is tractable.

Therefore, to prove (2), it will suffice to prove the following: For every cofibrant object $X \in \mathbf{Ind}^{\mathbf{S}}_\kappa(L(\mathcal{C}))$, the functor

$$X \otimes - \colon \mathbf{Ind}^{\mathbf{S}}_\kappa(L(\mathcal{C}))_{\mathrm{cof}} \to \mathbf{Ind}^{\mathbf{S}}_\kappa(L(\mathcal{C}))_{\mathrm{cof}}$$

preserves weak equivalences. But this is clear, because the composite

$$\mathrm{Fun}(\mathcal{C}^{\mathrm{op}}, \mathbf{S})_{\mathrm{cof}} \to \mathrm{Fun}(\mathcal{C}^{\mathrm{op}}, \mathcal{S}) \to \mathrm{Fun}(L(\mathcal{C})^{\mathrm{op}}, \mathcal{S}) \to \mathrm{Ind}_\kappa(L(\mathcal{C}))$$

is symmetric monoidal.

Part (3) follows from the fact that its right adjoint is the evaluation at the unit object, which is evidently right Quillen. The claim follows. □

Notation 4.5. For each replete subcategory[4] $\mathcal{X} \subset \mathcal{SMC}\mathsf{at}_\infty$, we will write $\mathsf{SMRelCat}_{\mathcal{X}} = \mathcal{X} \times_{\mathcal{SMC}\mathsf{at}_\infty} \mathsf{SMRelCat}$, and define $\mathsf{SMRelCat}_{(2,1),\mathcal{X}} \subset \mathsf{SMRelCat}_{(2,1)}$ for the sub $(2,1)$-category whose mapping groupoids are the components corresponding to the morphisms in $\mathsf{SMRelCat}_{\mathcal{X}}$.

We can now state the main result of this subsection.

Proposition 4.6. *Let κ be an uncountable regular cardinal. The strictly unitary pseudofunctor*

$$\begin{aligned} \mathsf{SMRelCat}_{\mathcal{SMC}\mathsf{at}_\infty(\kappa)} &\to \widehat{\mathsf{SMRelCat}}_{\kappa\text{-}\mathcal{PrSM},(2,1)} \\ \mathcal{C} &\mapsto \mathbf{Ind}_\kappa^{\mathbf{S}}(L(\mathcal{C}))_\flat \end{aligned}$$

induces a categorical equivalence

$$\mathsf{SMRelCat}_{\mathcal{SMC}\mathsf{at}_\infty(\kappa)}[\mathrm{loc.eq}^{-1}] \xrightarrow{\simeq} \widehat{\mathsf{SMRelCat}}_{\kappa\text{-}\mathcal{PrSM},(2,1)}[\mathrm{loc.eq}^{-1}].$$

We need a lemma for the proof of Proposition 4.6. We will use the following notation: For each replete subcategory $\mathcal{X} \subset \mathcal{SMC}\mathsf{at}_\infty$, we will write $\mathsf{SMRelCat}_{(2,1),\mathcal{X}} \subset \mathsf{SMRelCat}_{(2,1)}$ for the sub $(2,1)$-category whose mapping groupoids are the union of the components of the mapping groupoids of $\mathsf{SMRelCat}_{(2,1)}$ corresponding to the morphisms in $\mathsf{SMRelCat}_{\mathcal{X}}$. We then have:

Lemma 4.7. *For every replete subcategory $\mathcal{X} \subset \mathcal{SMC}\mathsf{at}_\infty$, the inclusion $\mathsf{SMRelCat}_{\mathcal{X}} \hookrightarrow \mathsf{SMRelCat}_{(2,1),\mathcal{X}}$ is a localization.*

Proof. According to [ACK25, Corollary 3.15], it suffices to show that the $(2,1)$-category $\mathsf{SMRelCat}_{(2,1),\mathcal{X}}$ admits a weak cotensor by [1]. Given an object $\mathcal{C} \in \mathsf{SMRelCat}_{(2,1)}$, a weak cotensor by [1] is given by $\mathcal{C}^J$, where the tensor product is given by the degreewise tensor product, and the weak equivalences are the natural transformations whose components are weak equivalences. □

We can now prove Proposition 4.6.

Proof of Proposition 4.6. The Yoneda embedding determines a pseudonatural transformation depicted as

$$\begin{tikzcd} \mathsf{SMRelCat}_{\mathcal{SMC}\mathsf{at}_\infty(\kappa)} \arrow[rr, "\mathbf{Ind}_\kappa^{\mathbf{S}}(L(-))_\flat"] \arrow[dr, hook] & \arrow[d, Rightarrow] & \widehat{\mathsf{SMRelCat}}_{\kappa\text{-}\mathcal{PrSM},(2,1)} \arrow[dl] \\ & \widehat{\mathsf{SMRelCat}}_{(2,1)} & \end{tikzcd}$$

Using Lemmas 3.1 and 4.7 to identify the localization of each category at local equivalences, we obtain a functor $I\colon \mathcal{SMC}\mathsf{at}_\infty(\kappa) \to \kappa\text{-}\mathcal{PrSM}$ and a natural transformation depicted as

$$\begin{tikzcd} \mathcal{SMC}\mathsf{at}_\infty(\kappa) \arrow[rr, "I"] \arrow[dr, hook] & \arrow[d, Rightarrow, "\alpha"] & \kappa\text{-}\mathcal{PrSM} \arrow[dl, hook] \\ & \widehat{\mathcal{SMC}\mathsf{at}}_\infty. & \end{tikzcd}$$

[4]Recall that a subcategory $\mathcal{C}' \subset \mathcal{C}$ of an ∞-category is called **replete** if every equivalence $f\colon X \xrightarrow{\simeq} X'$ in $\mathcal{C}$ such that $X' \in \mathcal{C}'$ is a morphism of $\mathcal{C}'$.

By construction, the pair (I, α) satisfies the hypothesis of Corollary B.13. It follows from this corollary that I is an equivalence, and we are done. □

4.2. **Strongly κ-combinatorial model categories.** Combinatorial model categories enjoy the following curious property, first articulated by Dugger in [Dug01a] and later expanded by Low [Low16]: Beyond sufficiently large regular cardinals, the distinction between ordinary-categorical notion and ∞-categorical notions gets blurry. In this subsection, we record several results that embody this principle.

The following definition is essentially due to Low [Low16, Definition 5.1]:

Definition 4.8. Let κ be a regular cardinal. We say that a model category $\mathbf{M}$ is **strongly κ-combinatorial** if there is a regular cardinal $\kappa_0 \lhd \kappa$ with the following properties:

(1) $\mathbf{M}$ is locally κ_0-presentable.
(2) $\mathbf{M}_\kappa$ is closed under finite limits in $\mathbf{M}$.
(3) Each hom-set in $\mathbf{M}_{\kappa_0}$ is κ-small.
(4) There are κ-small sets of morphisms in $\mathbf{M}_{\kappa_0}$ that cofibrantly generate the model structure on $\mathbf{M}$.

Remark 4.9. We can also define **strongly κ-tractable model categories** as strongly κ-combinatorial model categories satisfying conditions (1) through (4) of Definition 4.3 and the following additional condition: In (4), the elements in the generating sets must have cofibrant domain. Likewise, we have the notion of **strongly κ-tractable semi-model categories**.

The following results summarize the basic properties of strongly κ-combinatorial model categories.

Proposition 4.10. [Low16, Propositions 5.6, 5.12] *For every combinatorial model category $\mathbf{M}$, there is a regular cardinal κ such that $\mathbf{M}$ is strongly κ-combinatorial. Moreover, in this situation, $\mathbf{M}_\kappa$ is a model category whose cofibrations, fibrations, and weak equivalences are precisely those of $\mathbf{M}$ of κ-compact objects.*

Proposition 4.11. [Low16, Remark 5.2] *Let $\kappa' \rhd \kappa$ be regular cardinals. Every strongly κ-combinatorial model category is strongly κ'-combinatorial.*

Proposition 4.12. *Let κ be a regular cardinal, and let $\mathbf{M}$ be a strongly κ-combinatorial model category. Then the ∞-category $\mathbf{M}_{\kappa,\infty} = (\mathbf{M}_\kappa)_\infty$ admits κ-small colimits, and the functor*

$$\iota \colon \mathbf{M}_{\kappa,\infty} \to \mathbf{M}_\infty$$

exhibits $\mathbf{M}_\infty$ as an Ind_κ-completion of $\mathbf{M}_{\kappa,\infty}$.

Proof. The fact that $\mathbf{M}_{\kappa,\infty}$ admits κ-small colimits follows from the argument of [Cis19a, Proposition 7.7.4]. Next, to show that ι induces a categorical equivalence $\mathrm{Ind}_\kappa(\mathbf{M}_{\kappa,\infty}) \xrightarrow{\simeq} \mathbf{M}_\infty$, we use [Lur09, Proposition 5.3.5.11]. We must prove the following:

(1) The functor ι is fully faithful.
(2) The essential image of ι generates $\mathbf{M}_\infty$ under κ-filtered colimits.
(3) The image of ι lies in $(\mathbf{M}_\infty)_\kappa$.

Claim (1) follows from Proposition 4.10 and [ACK25, Corrollary 3.1], which asserts that the classical derived mapping spaces (computed using simplicial resolutions objects) computes the mapping spaces of the underlying ∞-categories of model categories.

For (2), recall that κ-filtered colimits in $\mathbf{M}$ are already homotopy colimits (Corollary A.16). Thus, we only have to show that $\mathbf{M}_\kappa$ generates $\mathbf{M}$ under κ-filtered colimits. This is clear, because $\mathbf{M}$ is locally κ-presentable.

For (3), suppose we are given a κ-filtered ∞-category $\mathcal{I}$ and a colimit diagram $\overline{F}\colon \mathcal{I}^{\triangleright} \to \mathbf{M}_\infty$. We must show that, for every object $X \in \mathbf{M}_\kappa$, the diagram $\mathbf{M}_\infty(X, -) \circ \overline{F}\colon \mathcal{I}^{\triangleright} \to \mathcal{S}$ is a colimit diagram. Using [Lur25, Tag 02QA], we may assume that $\mathcal{I}$ is (the nerve of) a poset. In this case, the functor $\mathrm{Fun}(\mathcal{I}^{\triangleright}, \mathbf{M}) \to \mathrm{Fun}(\mathcal{I}^{\triangleright}, \mathbf{M}_\infty)$ is a localization [Cis19a, Theorem 7.9.8], so we may assume that F lifts to a diagram $\overline{G}\colon \mathcal{I}^{\triangleright} \to \mathbf{M}$. Without loss of generality, we may assume that $\overline{G}$ takes values in the full subcategory of fibrant objects. Set $G = \overline{G}|\mathcal{I}$. Since $\mathcal{I}$ is κ-filtered, the map

$$\mathrm{colim}_{\mathcal{I}}\, G \to \overline{G}(\infty)$$

is a weak equivalence. Moreover, since fibrations of $\mathbf{M}$ are stable under κ-filtered colimits, the object $\mathrm{colim}_{\mathcal{I}}\, G \in \mathbf{M}$ is fibrant. Thus, we may assume that $\overline{G}$ is a strict colimit diagram.

Now choose a cosimplicial resolution $X^\bullet$ of X in $\mathbf{M}_\kappa$. By [ACK25, Corrollary 3.1], the functor $\mathbf{M}(X^\bullet, -)\colon \mathbf{M}_{\mathrm{fib}} \to \mathsf{sSet}$ descends to the functor $\mathbf{M}_\infty \to \mathcal{S}$ corepresented by X. Therefore, we are reduced to showing that the diagram $\mathbf{M}\big(X^\bullet, \overline{G}\big)\colon \mathcal{I}^{\triangleright} \to \mathsf{sSet}$ is a homotopy colimit diagram. Since $X^\bullet$ carries each object to a κ-compact object, this diagram is a colimit diagram. Since filtered colimits are homotopy colimits in sSet, we are done. □

Corollary 4.13. *Let κ be a regular cardinal, and let $\mathbf{M}$ be a symmetric $\mathbf{S}$-algebra whose underlying model category is strongly κ-combinatorial. The inclusion $\mathbf{M}_{\kappa,\mathrm{cof}} \hookrightarrow \mathbf{M}$ induces a left Quillen equivalence of $\mathbf{S}$-algebras*

$$\theta\colon \mathbf{Ind}_\kappa^{\mathbf{S}}(L(\mathbf{M}_{\kappa,\mathrm{cof}})) \xrightarrow{\simeq} \mathbf{M}$$
$$X \mapsto \int^{M \in \mathbf{M}_{\kappa,\mathrm{cof}}} M \otimes X(M).$$

Proof. We first show that θ induces a left Quillen functor

$$\theta'\colon \mathrm{Fun}\Big(\mathbf{M}_{\kappa,\mathrm{cof}}^{\mathrm{op}}, \mathbf{S}\Big) \to \mathbf{M},$$

where $\mathrm{Fun}\Big(\mathbf{M}_{\kappa,\mathrm{cof}}^{\mathrm{op}}, \mathbf{S}\Big)$ carries the projective model structure. Since $\mathbf{M}$ is strongly κ-combinatorial, it has a generating set I of cofibrations consisting of maps in $\mathbf{M}_\kappa$. The maps $\{\mathbf{M}(-, M) \otimes i\}_{M \in \mathbf{M}_{\kappa,\mathrm{cof}},\, i \in I}$ generate projective cofibrations of $\mathrm{Fun}\Big(\mathbf{M}_{\kappa,\mathrm{cof}}^{\mathrm{op}}, \mathbf{S}\Big)$ [Hir03, Theorem 11.6.1]. The images of these maps under θ are simply the maps $\{M \otimes i\}_{M \in \mathbf{M}_{\kappa,\mathrm{cof}},\, i \in I}$, which are all cofibrations. Hence θ carries projective cofibrations to cofibrations. A similar argument shows that θ carries projective trivial cofibrations to trivial cofibrations. Hence θ' is left Quillen, as claimed.

We now consider the following diagram, which commutes up to natural equivalence:

$$\begin{tikzcd}
\mathbf{M}_{\kappa,\mathrm{cof}} \arrow[r] \arrow[d] & \mathbf{M}_{\kappa,\infty} \arrow[d, "\iota"] \\
\mathrm{Fun}(\mathbf{M}_{\kappa,\mathrm{cof}}^{\mathrm{op}}, \mathbf{S})_\infty \arrow[r, "\theta'_\infty"] \arrow[d] & \mathbf{M}_\infty \\
\mathbf{Ind}_\kappa^{\mathbf{S}}(L(\mathbf{M}_{\kappa,\mathrm{cof}}))_\infty. &
\end{tikzcd}$$

According to Proposition 4.12, the functor ι exhibits $\mathbf{M}_\infty$ as an Ind_κ-completion of $\mathbf{M}_{\kappa,\infty}$. Also, by what we have just shown in the previous paragraph, the functor θ'_∞ preserves small colimits. It follows that θ'_∞ can be identified with the functor

$$\mathrm{Fun}\Big(\mathbf{M}_{\kappa,\mathrm{cof}}^{\mathrm{op}}, \mathcal{S}\Big) \to \mathrm{Ind}_\kappa(\mathbf{M}_{\kappa,\mathrm{cof}})$$

described in Construction 4.3. By the definition of weak equivalences of $\mathbf{Ind}_\kappa^{\mathbf{S}}(L(\mathbf{M}_{\kappa,\mathrm{cof}}))$, this means that θ preserves and reflects weak equivalences. Since we already know that θ' is a left Quillen functor, this is enough to conclude that θ is a left Quillen functor. The resulting functor

$$\theta_\infty \colon \mathbf{Ind}_\kappa^{\mathbf{S}}(L(\mathbf{M}_{\kappa,\mathrm{cof}}))_\infty \to \mathbf{M}_\infty$$

is an equivalence, because both sides are the localization of $\mathrm{Fun}\Big(\mathbf{M}_{\kappa,\mathrm{cof}}^{\mathrm{op}}, \mathbf{S}\Big)_\infty$ at the same class of morphisms. The proof is now complete. □

We conclude this subsection by taking monoidal structures into account.

Definition 4.14. Let κ be a regular cardinal. A **monoidally κ-combinatorial model category** is a monoidal model category $\mathbf{M}$ satisfying the following conditions:

- As a model category, $\mathbf{M}$ is strongly κ-combinatorial.
- κ-compact objects are stable under finite tensor products. (In particular, the unit object is κ-compact.)

We write $\mathsf{CombMMC}(\kappa) \subset \mathsf{CombMMC}$ for the subcategory spanned by the monoidally κ-combinatorial model category and those maps that preserve κ-compact objects.

Lemma 4.15.

(1) For every pair of regular cardinals $\kappa \lhd \lambda$, we have $\mathsf{CombMMC}(\kappa) \subset \mathsf{CombMMC}(\lambda)$.
(2) Every morphism of $\mathsf{CombMMC}$ belongs to $\mathsf{CombMMC}(\kappa)$ for some regular cardinal κ.

Proof. We first show that every combinatorial monoidal model category $\mathbf{M}$ is monoidally κ-combinatorial for some κ, and that such an $\mathbf{M}$ is monoidally λ-combinatorial for every $\lambda \rhd \kappa$. By Proposition 4.10, there is some regular cardinal κ_0 such that $\mathbf{M}$ is strongly κ_0-combinatorial as a model category. Find a regular cardinal $\kappa \rhd \kappa_0$ such that, if $X, Y \in \mathbf{M}_{\kappa_0}$, then $X \otimes Y \in \mathbf{M}_\kappa$. Since every κ-compact object is a κ-small colimit of κ_0-compact object [MP89, Theorem 2.3.11], this ensures that κ-compact objects is stable under tensor product. Hence, $\mathbf{M}$ is monoidally κ-combinatorial. A similar argument, using Proposition 4.11, shows that every monoidally κ-combinatorial model category is monoidally λ-combinatorial.

We now prove (1) and (2). Part (1) follows from the result in the previous paragraph and [AR94, Remark 2.20]. For (2), let $F \colon \mathbf{M} \to \mathbf{N}$ be a morphism of $\mathsf{CombMMC}$. By the result in the previous paragraph and part (1), we can find some κ_0 such that $\mathbf{M}$ and $\mathbf{N}$ are monoidally κ_0-combinatorial. Choose $\kappa \rhd \kappa_0$ such that $F(\mathbf{M}_{\kappa_0}) \subset \mathbf{N}_\kappa$. We claim that F belongs to $\mathsf{CombMMC}(\kappa)$.

Since every κ-compact object is a κ-small colimit of κ_0-compact object [MP89, Theorem 2.3.11], our choice of κ and κ_0 ensures that $F(\mathbf{M}_\kappa) \subset \mathbf{N}_\kappa$. We also know from (1) that $\mathbf{M}$ and $\mathbf{N}$ are monoidally κ-combinatorial. Therefore, F belongs to $\mathsf{CombMMC}(\kappa)$, as desired. □

4.3. **Proof of Proposition 2.6.** We now arrive at the proof of Proposition 2.6.

Notation 4.16. We write urCard for the (large) poset of (small) uncountable regular cardinals, ordered by $\lhd$. Given a functor $F \colon \mathsf{urCard} \to \widehat{\mathsf{Cat}}$, we write $\int^{\mathsf{urCard}} F \to \mathsf{urCard}$ for the Grothendieck construction of F. Likewise, given a functor $G \colon \mathsf{urCard}^{\mathrm{op}} \to \widehat{\mathsf{Cat}}$, we write $\int_{\mathsf{urCard}} G \to \mathsf{urCard}$ for its Grothendieck construction.

Notation 4.17. We let $\mathsf{combsym}\mathbf{S}\text{-}\mathsf{alg} \subset \mathsf{CombSym}\mathbf{S}\text{-}\mathsf{Alg}$ denote the full subcategory of $\mathbf{S}$-algebras $\mathbf{M}$ with the following property: For every regular cardinal κ

such that (the underlying category of) $\mathbf{M}$ is κ-presentable, the full subcategory $\mathbf{M}_\kappa \subset \mathbf{M}$ of κ-compact objects is *literally* small (not just esssentially small).

We define a full subcategory $\mathsf{combsym}\mathbf{S}\text{-}\mathsf{alg}_{(2,1)} \subset \mathsf{combsym}\mathbf{S}\text{-}\mathsf{alg}_{(2,1)}$ similarly.

Remark 4.18. The functor $\mathsf{combsym}\mathbf{S}\text{-}\mathsf{alg} \to \mathsf{combsym}\mathbf{S}\text{-}\mathsf{alg}_{(2,1)}$ is a localization. This follows from the argument in Proposition 1.8 and the following observation [Low13, Proposition 2.23]: Let κ be a regular cardinal, and let $\mathbf{C}$ be a locally κ-presentable category. For every category $\mathbf{I}$ with only finitely many morphisms, the category $\mathrm{Fun}(\mathbf{I}, \mathbf{C})$ is locally κ-presentable, and its κ-compact objects are exactly those diagrams $\mathbf{I} \to \mathbf{C}$ carrying each object to a κ-compact object.

Proof of Proposition 2.6. Consider the following diagram:

$$\begin{tikzcd}
\mathsf{combsym}\mathbf{S}\text{-}\mathsf{alg}[\mathrm{Quill.eq}^{-1}] \arrow[r] \arrow[d, "\alpha'"] & \mathsf{CombSym}\mathbf{S}\text{-}\mathsf{Alg}[\mathrm{Quill.eq}^{-1}] \arrow[r] \arrow[d, "\alpha"'] & \widehat{\mathsf{SMRelCat}}_{\mathcal{P}\mathrm{r}\mathcal{SM}} \arrow[d, "\beta"'] \\
\mathsf{combsym}\mathbf{S}\text{-}\mathsf{alg}_{(2,1)}[\mathrm{Quill.eq}^{-1}] \arrow[r, "\gamma"] & \mathsf{CombSym}\mathbf{S}\text{-}\mathsf{Alg}_{(2,1)}[\mathrm{Quill.eq}^{-1}] \arrow[r] & \widehat{\mathsf{SMRelCat}}_{\mathcal{P}\mathrm{r}\mathcal{SM},(2,1)}
\end{tikzcd}$$

By Proposition 1.8, the map α is a localization, and we saw in Remark 4.18 that α' is also a localization. The map β is a localization by Lemma 4.7. Furthermore, the map γ is an equivalence (as it is an equivalence even before localizing). Thus, it suffices to show that bottom composite

$$\mathsf{combsym}\mathbf{S}\text{-}\mathsf{alg}_{(2,1)}[\mathrm{Quill.eq}^{-1}] \to \widehat{\mathsf{SMRelCat}}_{(2,1)}[\mathrm{loc.eq}^{-1}]$$

is an equivalence.

For each uncountable regular cardinal κ, let $\mathsf{combsym}\mathbf{S}\text{-}\mathsf{alg}(\kappa)$ denote the fiber product

$$\mathsf{combsym}\mathbf{S}\text{-}\mathsf{alg} \times_{\mathsf{CombMMC}} \mathsf{CombMMC}(\kappa).$$

We now consider the following diagram:

$$\begin{tikzcd}
\int^{\mathsf{urCard}} \mathsf{combsym}\mathbf{S}\text{-}\mathsf{alg}(-) \arrow[r, "F"] \arrow[d, "U"'] & \int_{\mathsf{urCard}} \mathsf{SMRelCat}_{\mathcal{SM}\mathcal{C}\mathsf{at}_\infty(-)} \arrow[dl, "\mathbf{Ind}^{\mathbf{S}}"'] \arrow[d, "(-)_\flat \circ \mathbf{Ind}^{\mathbf{S}}"] \\
\mathsf{combsym}\mathbf{S}\text{-}\mathsf{alg}_{(2,1)} \arrow[r, "(-)_\flat"'] & \widehat{\mathsf{SMRelCat}}_{\mathcal{P}\mathrm{r}\mathcal{SM},(2,1)}
\end{tikzcd}$$

Here U and F are functors and $\mathbf{Ind}^{\mathbf{S}}$ is a strictly unitary pseudofunctor, defined by the formulas

$$\begin{aligned}
F(\kappa, \mathbf{M}) &= (\kappa, \mathbf{M}_{\kappa,\mathrm{cof}}), \\
U(\kappa, \mathbf{M}) &= \mathbf{M}, \\
\mathbf{Ind}^{\mathbf{S}}(\kappa, \mathcal{C}) &= \mathbf{Ind}^{\mathbf{S}}_\kappa(L(\mathcal{C})).
\end{aligned}$$

The lower triangle commutes by construction; the upper triangle does not commute on the nose, but Corollary 4.13 gives a pseudonatural Quillen equivalence $\mathbf{Ind}^{\mathbf{S}} \circ F \xrightarrow{\simeq} U$. Thus, to prove the claim, it suffices to show that U and $(-)_\flat \circ \mathbf{Ind}^{\mathbf{S}}$ induce equivalences of ∞-categories when localized at the maps whose images in $\mathcal{P}\mathrm{r}\mathcal{SM}$ are equivalences.

We start from the claim on U. We claim more strongly that U is a localization. According to Lemma 4.15, the category $\mathsf{combsym}\mathbf{S}\text{-}\mathsf{alg}$ is the colimit of $\mathsf{combsym}\mathbf{S}\text{-}\mathsf{alg}(\kappa)$ as κ ranges over urCard. It follows from [Lur25, Tag 02UU] that the functor

$$\int^{\mathsf{urCard}} \mathsf{combsym}\mathbf{S}\text{-}\mathsf{alg}(-) \to \mathsf{combsym}\mathbf{S}\text{-}\mathsf{alg}$$

is a localization. Since the functor $\mathsf{combsym}\mathbf{S}\text{-}\mathsf{alg} \to \mathsf{combsym}\mathbf{S}\text{-}\mathsf{alg}_{(2,1)}$ is also a localization (Remark 4.18), we deduce that U is also a localization.

Next, for $(-)_\flat \circ \mathbf{Ind}^{\mathbf{S}}$, we factor it as

$$\int_{\mathsf{urCard}} \mathsf{SMRelCat}_{\mathcal{SMC}\mathsf{at}_\infty(-)} \xrightarrow{\Phi} \int_{\mathsf{urCard}} \widehat{\mathsf{SMRelCat}}_{(-)\text{-}\mathcal{PrSM},(2,1)} \xrightarrow{\Psi} \widehat{\mathsf{SMRelCat}}_{\mathcal{PrSM},(2,1)},$$

where $\int^{\mathsf{urCard}} \widehat{\mathsf{SMRelCat}}_{(-)\text{-}\mathcal{PrSM},(2,1)}$ denotes the cartesian fibration associated with the functor $\kappa \mapsto \widehat{\mathsf{SMRelCat}}_{\kappa\text{-}\mathcal{PrSM},(2,1)}$, concretely realized as (the dual of) Lurie's relative nerve [Lur09, Definition 3.2.5.2]. The functor Φ is induced by the functors $\{\mathbf{Ind}^{\mathbf{S}}_\kappa(L(-))_\flat\}_\kappa$, and Ψ is the forgetful functor. As in the previous paragraph, the functor Ψ is already a localization, so it suffices to verify that Φ induces a localization upon localizing at the maps whose images in $\mathcal{PrSM}$ are equivalences. This follows from (the dual of) [Lur25, Tag 02LW] and Proposition 4.6, which show more strongly that it induces a localization when we localize at fiberwise local equivalences. The proof is now complete. □

5. Proof of Proposition 2.7

In this section, we take up the proof of Proposition 2.7. In light of Propositions 2.5 and 2.6, it will suffice to prove this for one *specific* choice of $\mathbf{S}$. To this end, we will construct a new model category of *symmetric cubical sets* in Subsection 5.1. We then use this to prove Proposition 2.7 in Subsection 5.2.

5.1. Symmetric cubical sets.

In this subsection, we construct the symmetric monoidal model category of symmetric cubical sets (Theorem 5.9). It has two essential features: Firstly, it models the homotopy theory of spaces, in the sense that its underlying symmetric monoidal ∞-category is equivalent to the cartesian monoidal ∞-category of spaces. Secondly, every combinatorial symmetric monoidal model category with cofibrant unit can be enriched over symmetric cubical sets in an essentially unique manner (Corollary 5.11). Together, these features make the category of symmetric cubical sets an ideal choice of $\mathbf{S}$.

Remark 5.1. Part of the contents of this subsection is similar in spirit to Isaacson's papers [Isa11, Isa09]. In these papers, he considered symmetric cubical sets by using a slightly bigger category of symmetric cubes. He then showed that every combinatorial symmetric monoidal model category with cofibrant unit and which satisfies the monoid axiom can be enriched over symmetric cubical sets [Isa09, Theorem 10.1]. However, his approach does not seem to give an essential uniqueness of this enrichment.

We start by recalling the classical cube category and the Grothendieck model structure.

Definition 5.2. The **box category** $\square$ has the following descriptions:

- Objects are the posets $[1]^n$, where $n \geq 0$.
- A morphism $f\colon [1]^n \to [1]^m$ is a poset map that erases coordinates and inserts 0 and 1 without changing the order of the coordinates. Equivalently, they are the generated under composition by the following maps:
  (1) The **face map** $\delta^{i,\varepsilon} = \delta^{i,\varepsilon}_n\colon [1]^n \to [1]^{n+1}$ for $n \geq 0$ and $1 \leq i \leq n+1$, which inserts $\varepsilon \in \{0,1\}$ in the ith coordinate.
  (2) The **degeneracy map** $\sigma^i_n\colon [1]^{n+1} \to [1]^n$ for $n \geq 0$ and $1 \leq i \leq n+1$, which deletes the ith coordinate.

We will write $\square_{\leq 1} \subset \square$ for the full subcategory spanned by the objects $[1]^n$ with $n \leq 1$.

The category $\square$ is naturally a Reedy category. (In fact, an Eilenberg–Zilber category in the sense of [Cis19a, Definition 1.3.1].) Also, the cartesian product of posets makes $\square$ into a monoidal category. The category $\mathsf{Set}^{\square^{\mathrm{op}}}$ of **cubical sets** admits an induced monoidal structure, given by Day's convolution product. The presheaf represented by $[1]^n$ will be denote by $\square^n$.

The category $\mathsf{Set}^{\square^{\mathrm{op}}}$ can be endowed with a model structure which models the homotopy theory of spaces. To state it more precisely, set $\partial\square^n = \bigcup_{i,\varepsilon} \delta^{i,\varepsilon}\big(\square^{n-1}\big)$ and $\sqcap^n_{i,\varepsilon} = \bigcup_{(j,\eta)\neq(i,\varepsilon)} \delta^{i,\varepsilon}\big(\square^{n-1}\big)$. We consider the following sets of morphisms of cubical sets:

$$I = \{\partial\square^n \to \square^n \mid n \geq 0\},\ J = \{\sqcap^n_{i,\varepsilon} \to \square^n \mid n \geq 1,\ 1 \leq i \leq n,\ \varepsilon \in \{0,1\}\}.$$

Theorem 5.3. [Cis06], [Jar06, Theorem 6.2, Theorem 8.6, Theorem 8.8] *The category $\mathsf{Set}^{\square^{\mathrm{op}}}$ of **cubical sets** has a combinatorial monoidal model structure called the **Grothendieck model structure**, whose:*

- *cofibrations are the monomorphisms.*
- *weak equivalences are preserved and detected by the triangulation functor $\mathsf{Set}^{\square^{\mathrm{op}}} \to \mathsf{Set}^{\boldsymbol{\Delta}^{\mathrm{op}}}$, which is the left adjoint carrying $\square^n \mapsto \big(\Delta^1\big)^n$.*

Moreover, I and J cofibrantly generate this model structure, and T is a left Quillen equivalence with respect to the Kan–Quillen model structure on simplicial sets.

Having recalled the basics of cubical sets, we now develop the theory of symmetric cubical sets. Our goal is to consider the symmetric monoidal version of Theorem 5.3 (Theorem 5.9) and to prove its model-categorical universal property (Corollary 5.11).

Definition 5.4. We define the symmetric monoidal category of **symmetric box category** $\square_\Sigma$ by adjoining to the box category the permutation map $\pi_p\colon [1]^n \to [1]^n$ for each $n \geq 0$ and $p \in \Sigma_n$, defined by $\pi_p(x_1,\dots,x_n) = \big(x_{p^{-1}(1)},\dots,x_{p^{-1}(n)}\big)$. There is an inclusion $i\colon \square \to \square_\Sigma$. We denote by $\square^n_\Sigma$ the presheaves on $\square_\Sigma$ represented by $[1]^n$. Note that $\square_{\leq 1}$ is a full subcategory of $\square_\Sigma$.

The following proposition is immediate from the definitions:

Proposition 5.5. *We have the following **cocubical relations** for maps in $\square_\Sigma$:*

- $\delta^{j,\eta}\delta^{i,\varepsilon} = \delta^{i,\varepsilon}\delta^{j-1,\eta}$ *for* $j \leq i$.
- $\sigma^j\delta^{i,\varepsilon} = \begin{cases} \delta^{i,\varepsilon}\sigma^{j-1} & \text{if } i < j, \\ \mathrm{id} & \text{if } i = j, \\ \delta^{i-1}\sigma^j & \text{if } i > j. \end{cases}$
- $\sigma^j\sigma^i = \sigma^i\sigma^{j+1}$ *if* $i \leq j$.
- $\pi_p\pi_q = \pi_{pq}$.
- $\pi_p\delta^{i,\varepsilon} = \delta^{p(i)}\pi_q$, *where q denotes the composite*

$$\{1,\dots,n\} \cong \{1,\dots,n+1\} \setminus \{i\} \xrightarrow{p} \{1,\dots,n+1\} \setminus \{p(i)\} \cong \{1,\dots,n\}.$$

- $\sigma^i\pi_p = \pi_q\sigma^{p^{-1}(i)}$, *where q denotes the composite*

$$\{1,\dots,n-1\} \cong \{1,\dots n\} \setminus \{p^{-1}(i)\} \xrightarrow{p} \{1,\dots,n\} \setminus \{i\} \cong \{1,\dots,n-1\}.$$

Moreover, every map in $\square_\Sigma$ can be written uniquely as

$$\delta^{i_1,\varepsilon_1}\cdots\delta^{i_n\varepsilon_n}\pi_p\sigma^{j_1}_l\cdots\sigma^{j_m},$$

where $i_1 > \cdots i_n$, $j_1 < \cdots < j_m$, and $p \in \Sigma_l$.

Corollary 5.6. *The symmetric box category $\square_\Sigma$ is freely generated by the face maps, degeneracy maps, and permutation maps, subject to the cocubical relations. A similar claim holds for the classical box category.*

Corollary 5.6 is quite elementary, but it has an important consequence. Namely, it can be used to show that $\square_\Sigma$ is the free symmetric monoidal category generated by an "interval":

Corollary 5.7. *Let $(\mathcal{C}, \otimes, \mathbf{1})$ be a symmetric monoidal category. The evaluation at the unit object $[1] \in \square_\Sigma$ determines a categorical equivalence*

$$\theta\colon \operatorname{Fun}^{\otimes}(\square_\Sigma, \mathcal{C}) \xrightarrow{\simeq} \operatorname{Fun}(\square_{\leq 1}, \mathcal{C}) \times_{\operatorname{Fun}(\{[1]^0\},\mathcal{C})} \{\mathbf{1}\}.$$

A similar claim holds for monoidal categories and the plain cube category.

Proof. The functor θ needs a bit of explanation. If $F\colon \square_\Sigma \to \mathcal{C}$ is a symmetric monoidal functor, then its image in $\operatorname{Fun}(\square_{\leq 1}, \mathcal{C}) \times_{\operatorname{Fun}(\{[1]^0\},\mathcal{C})} \{\mathbf{1}\}$ is given by the functor $\square_{\leq 1} \to \mathcal{C}$ obtained by modifying the value of F at $[1]^0$ to $\mathbf{1}$ using the structure map $\mathbf{1} \xrightarrow{\cong} F\big([1]^0\big)$ of F.

With this in mind, we will give an explicit inverse equivalence of θ. Given a functor $F\colon \square_{\leq 1} \to \mathcal{C}$ carrying $[1]^0$ to the unit object, we can define a new functor $\widetilde{F}\colon \square_\Sigma \to \mathcal{C}$ by $\widetilde{F}([1]^n) = F([1])^{\otimes n}$, with structure maps given by that of F and the braiding of $\mathcal{C}$ (Corollary 5.6). The coherence maps of $\mathcal{C}$ makes $\widetilde{F}$ into a symmetric monoidal functor. The functor $F \mapsto \widetilde{F}$ is an inverse equivalence of θ. □

Corollary 5.8. *Let $(\mathcal{C}, \otimes, \mathbf{1})$ be a cocomplete symmetric monoidal category whose tensor product preserves small colimits in each variable. The evaluation at the unit object $\square_\Sigma^0 \in \mathsf{Set}^{\square_\Sigma^{\mathrm{op}}}$ induces a categorical equivalence*

$$\operatorname{Fun}^{\otimes,cc}\Big(\mathsf{Set}^{\square_\Sigma^{\mathrm{op}}}, \mathcal{C}\Big) \xrightarrow{\simeq} \operatorname{Fun}(\square_{\leq 1}, \mathcal{C}) \times_{\operatorname{Fun}(\{[1]^0\},\mathcal{C})} \{\mathbf{1}\},$$

where $\operatorname{Fun}^{\otimes,cc}\Big(\mathsf{Set}^{\square_\Sigma^{\mathrm{op}}}, \mathcal{C}\Big)$ denotes the category of symmetric monoidal categories whose underlying functor preserves small colimits, and symmetric monoidal natural transformations between them.

Proof. This follows from Corollary 5.7 and the universal property of Day convolution product monoidal structure. □

We can now state our first main result of this section. For the statement theorem, we will write $\partial\square_\Sigma^n \subset \square_\Sigma^n$ for the subpresheaf consisting of the maps $[1]^k \to [1]^n$ that inserts at least one 0 or 1. Equivalently, we have $i_!(\partial\square^n) = \partial\square_\Sigma^n$.

Theorem 5.9. *The category $\mathsf{Set}^{\square_\Sigma^{\mathrm{op}}}$ has a tractable model structure whose weak equivalences and fibrations are preserved and detected by the forgetful functor $i^*\colon \mathsf{Set}^{\square_\Sigma^{\mathrm{op}}} \to \mathsf{Set}^{\square^{\mathrm{op}}}$. Moreover:*

(I) The adjunction

$$i_!\colon \mathsf{Set}^{\square^{\mathrm{op}}} \underset{\longleftarrow}{\overset{\longrightarrow}{\perp}} \mathsf{Set}^{\square_\Sigma^{\mathrm{op}}} : i^*$$

is a Quillen equivalence.

(II) The model structure on $\mathsf{Set}^{\square_\Sigma^{\mathrm{op}}}$ is symmetric monoidal with respect to the Day convolution.

(III) The model structure is left proper.

(IV) If $\mathbf{M}$ is a symmetric monoidal model category and $F\colon \mathsf{Set}^{\square_\Sigma^{\mathrm{op}}} \to \mathbf{M}$ is a symmetric monoidal functor which is also a left adjoint, then F is left Quillen precisely when the following conditions are satisfied:

- *The map $F\big(\partial\square_\Sigma^1 \to \square_\Sigma^1\big)$ is a cofibration.*

- *The map* $F\big(\square^1_\Sigma\big) \to F\big(\square^0_\Sigma\big)$ *is a weak equivalence.*
- *The monoidal unit of* $\mathbf{M}$ *is cofibrant.*

(V) *The assignment* $\square^n_\Sigma \mapsto \big(\Delta^1\big)^n$ *determines a symmetric monoidal left Quillen equivalence* $\mathsf{Set}^{\square^{\mathrm{op}}_\Sigma} \to \mathsf{Set}^{\boldsymbol{\Delta}^{\mathrm{op}}}$.

The proof of Theorem 5.9 relies on the following lemma.

Lemma 5.10. *The diagram*

$$\begin{tikzcd} \mathsf{Set}^{\square^{\mathrm{op}}} \arrow[r, hook] \arrow[d] & \mathcal{S}^{\square^{\mathrm{op}}} \arrow[d, "\mathrm{colim}_{\square^{\mathrm{op}}}"] \\ \mathsf{Set}^{\square^{\mathrm{op}}}[\mathrm{weq}^{-1}] \arrow[r, "\simeq"] & \mathcal{S} \end{tikzcd}$$

commutes up to natural equivalence.

Proof. We first give a model for $\mathrm{colim}_{\square^{\mathrm{op}}}$. Consider the functor

$$\Phi\colon \mathrm{Fun}\Big(\square^{\mathrm{op}}, \mathsf{Set}^{\square^{\mathrm{op}}}\Big) \to \mathsf{Set}^{\square^{\mathrm{op}}},\ Y \mapsto \int^{[1]^n \in \square} Y_n \otimes \square^n.$$

Since the cocubical object $\square^\bullet \in \mathrm{Fun}\Big(\square, \mathsf{Set}^{\square^{\mathrm{op}}}\Big)$ is Reedy cofibrant, [Lur09, Proposition A.2.9.26] shows that the functor Φ is left Quillen for the Reedy model structure. The right adjoint of this functor, given by $K \mapsto K^{\square^\bullet}$, is weakly equivalent to that of the diagonal functor on the full subcategory of fibrant objects. Therefore, Φ is a model of the homotopy colimit, in the sense that the induced functor

$$\mathcal{S}^{\square^{\mathrm{op}}} \to \mathcal{S}$$

is equivalent to $\mathrm{colim}_{\square^{\mathrm{op}}}$.

Now since every cubical set X is Reedy cofibrant when regarded as a levelwise discrete cubical object in $\mathsf{Set}^{\square^{\mathrm{op}}}$, the above argument shows that the composite of the functor $\Phi'\colon \mathsf{Set}^{\square^{\mathrm{op}}} \to \mathsf{Set}^{\square^{\mathrm{op}}}$, $X \mapsto \Phi(X)$ and the localization functor $\mathsf{Set}^{\square^{\mathrm{op}}} \to \mathcal{S}$ is equivalent to the composite

$$\mathsf{Set}^{\square^{\mathrm{op}}} \to \mathcal{S}^{\square^{\mathrm{op}}} \to \mathcal{S}.$$

On the other hand, Φ' is naturally isomorphic to the identity functor by the coYoneda lemma. The claim follows. $\square$

Proof of Theorem 5.9. We start by showing that the model structure on $\mathsf{Set}^{\square^{\mathrm{op}}}$ transfers to a model structure on $\mathsf{Set}^{\square^{\mathrm{op}}_\Sigma}$, using [Hir03, Theorem 11.3.2]. Let $I = \{\partial\square^n \to \square^n \mid n \geq 0\}$ and $J = \{\sqcap^n_{i,\varepsilon} \to \square^n \mid n \geq 1,\ 1 \leq i \leq n,\ \varepsilon \in \{0,1\}\}$ denote the generating cofibrations and trivial cofibrations of $\mathsf{Set}^{\square^{\mathrm{op}}}$. We wish to show that i^* takes $i_!J$-cell complexes to weak equivalences. For this, it will suffice to show that each of the map in $i^*i_!J$ is a trivial cofibration. The cofibration part is clear, because $i_!$ and i^* preserve monomorphisms. Consequently, it will suffice to show that each of the maps in $i^*i_!J$ is a weak equivalence. We will prove this by showing that the unit $\eta\colon \mathrm{id} \to i^*i_!$ is a natural weak equivalence.

By the standard argument using the skeletal filtration of presheaves [Cis19a, Theorem 1.3.8], it will suffice to show that $\eta_{\square^n}\colon \square^n \to i^*i_!\square^n$ is a weak equivalence for all n. In other words, our task it to prove that each $i^*i_!\square^n$ is weakly contractible. By Lemma 5.10 and [Lur09, Corollary 3.3.4.6], this is equivalent to the condition that the category $\square_{/i^*i_!\square^n} \cong \square \times_{\square_\Sigma} \square_{\Sigma/\square^n_\Sigma}$ be weakly contractible for all n. Thus, it suffices to show that the functor $i\colon \square \to \square_\Sigma$ is homotopy initial.

The homotopy initiality of i is proved as follows: Consider the commutative diagram

$$\begin{tikzcd} & \square_+ \arrow[dl, "j"'] \arrow[dr, "k"] & \\ \square \arrow[rr, "i"'] & & \square_\Sigma, \end{tikzcd}$$

where $\square_+ \subset \square$ denotes the subcategory spanned by the face maps. Since homotopy initial functors have the right cancellation property [Cis19b, Corollary 4.1.9], it will suffice to show that j and k are homotopy initial. We will show that j is homotopy initial; the proof for k is similar. Our task is to show that the category $\square_+ \times_\square \square_{/\square^n}$ is weakly contractible for every $n \geq 0$. But the inclusion $\square_+ \times_\square \square_{/\square^n} \hookrightarrow \square_{/\square^n}$ admits a homotopy inverse on the level of classifying spaces, given by the functor that factors a map as a composition of degeneracy maps followed by a composition of face maps. So we are reduced to showing that $\square_{/\square^n}$ is weakly contractible, which is clear.

We now prove (I) through (V). For (I), observe that we have just shown that the derived unit is an isomorphism. Since $\mathbb{R}i^*$ is conservative by construction, the triangle identities show that the derived counit is also an isomorphism. Hence, the adjunction is a Quillen equivalence. Part (II) follows from the fact that the model structure on $\mathsf{Set}^{\square^{\mathrm{op}}}$ is monoidal for the Day convolution and that $i_!$ is a monoidal functor. Part (III) follows from the left properness of $\mathsf{Set}^{\square^{\mathrm{op}}}$, since i^* preserves cofibrations and small colimits and detects weak equivalences. Part (IV) is proved exactly as in [Law17, Corollary 1.5]. Finally, part (V) follows from parts (I, IV) and Theorem 5.3. $\square$

Combining Corollary 5.8 and Theorem 5.9, we get the following model-categorical universal property of symmetric cubical sets:

Corollary 5.11. *Every cofibrantly generated symmetric monoidal model category* $\mathbf{M}$ *with a cofibrant unit is a symmetric* $\mathsf{Set}^{\square^{\mathrm{op}}_\Sigma}$*-algebra in an essentially unique way. More precisely, the category* $\mathrm{Fun}^{\otimes, LQ}\left(\mathsf{Set}^{\square^{\mathrm{op}}_\Sigma}, \mathbf{M}\right)$ *is weakly contractible.*

Proof. Define a full subcategory $\mathcal{X}$ of

$$\mathrm{Fun}(\square_{\leq 1}, \mathbf{M})_{\mathbf{1}} = \mathrm{Fun}(\square_{\leq 1}, \mathbf{M}) \times_{\mathrm{Fun}(\{[1]^0\}, \mathbf{M})} \{\mathbf{1}\}$$

as follows: An object of $F \in \mathrm{Fun}(\square_{\leq 1}, \mathbf{M})_{\mathbf{1}}$ can be identified with a diagram of the form

$$\begin{tikzcd} \mathbf{1} \arrow[dr] \arrow[drr, bend left, "\mathrm{id}"] & & \\ & F([1]) \arrow[r] & \mathbf{1}. \\ \mathbf{1} \arrow[ur] \arrow[urr, bend right, "\mathrm{id}"'] & & \end{tikzcd}$$

We declare that such a diagram belongs to $\mathcal{X}$ if and only if the map $\mathbf{1} \amalg \mathbf{1} \to F([1])$ is a cofibration and the map $F([1]) \to \mathbf{1}$ is a weak equivalence.

According to Corollary 5.8 and Theorem 5.9, there is an equivalence of categories

$$\mathrm{Fun}^{\otimes, LQ}\left(\mathsf{Set}^{\square^{\mathrm{op}}_\Sigma}, \mathbf{M}\right) \xrightarrow{\simeq} \mathcal{X}.$$

Therefore, it suffices to show that $\mathcal{X}$ is weakly contractible. To this end, we define a functor $\Phi\colon \mathrm{Fun}(\square_{\leq 1}, \mathbf{M})_{\mathbf{1}} \to \mathrm{Fun}(\square_{\leq 1}, \mathbf{M})_{\mathbf{1}}$ as follows: It carries an object $(\mathbf{1} \amalg \mathbf{1} \to I \to \mathbf{1})$ to an object $(\mathbf{1} \amalg \mathbf{1} \to I' \to \mathbf{1})$, where I' is obtained by (functorially) factoring the map $\mathbf{1} \amalg \mathbf{1} \to I$ as a cofibration $\mathbf{1} \amalg \mathbf{1} \rightarrowtail I'$ followed by a

weak equivalence $I' \xrightarrow{\simeq} I$. Then Φ restricts to a functor $\Phi_{\mathcal{X}} \colon \mathcal{X} \to \mathcal{X}$. The functor $\Phi_{\mathcal{X}}$ admits natural transformations to the identity functor and the constant functor at $\Phi(\mathbf{1} \amalg \mathbf{1} \to \mathbf{1} \to \mathbf{1})$. Therefore, the identity functor of $\mathcal{X}$ can be connected by a zig-zag of natural transformation to a constant functor. Hence $\mathcal{X}$ is weakly contractible, as claimed. □

5.2. **Proof of Proposition 2.7.** We can now give a proof of Proposition 2.7.

Proof of Proposition 2.7. As discussed at the beginning of the section, it will suffice to prove the claim for a specific choice of $\mathbf{S}$.

Take $\mathbf{S} = \mathsf{Set}^{\square^{\mathrm{op}}_{\Sigma}}$. The functor $U \colon \mathsf{CombSym}\mathbf{S}\text{-}\mathsf{Alg} \to \mathsf{CombSMMC}^{\mathbf{1}}$ is a cocartesian fibration, and its fibers are weakly contractible by Corollary 5.11. It follows from [Lur25, Tag 02LY] that U is already a localization. In particular, it induces an equivalence

$$\mathsf{CombSym}\mathbf{S}\text{-}\mathsf{Alg}[\mathrm{Quill.eq}^{-1}] \xrightarrow{\simeq} \mathsf{CombSMMC}^{\mathbf{1}}[\mathrm{Quill.eq}^{-1}],$$

as claimed. □

6. Variations of the main theorem

Notation 6.1. Throughout this section, we fix a left proper tractable symmetric monoidal model category $\mathbf{S}$ with a cofibrant unit, whose underlying symmetric monoidal ∞-category is equivalent to the cartesian monoidal ∞-category $\mathcal{S}$ of ∞-groupoids.

In Section 1, we introduced many variations of categories of symmetric monoidal model categories. In this section, we will identify the localizations of most of them. We also identify the localizations of their subcategories of left Quillen equivalences.

This section has two main results, one in the symmetric monoidal case (Theorem 6.2), and the other in the non-symmetric monoidal case (Theorem 6.4). Let us start by stating the symmetric monoidal case.

Theorem 6.2. *Consider the diagram of categories introduced in Section 1:*

$$\begin{tikzcd}
 & \mathsf{TractSMMC}_{\mathbf{S},\mathrm{semi}} \arrow[dr] & \\
\mathsf{TractSMMC}_{\mathbf{S}} \arrow[ur] \arrow[r] \arrow[d] & \mathsf{TractSMMC} \arrow[r] \arrow[d] & \mathsf{TractSMMC}_{\mathrm{semi}} \\
\mathsf{CombSMMC}_{\mathbf{S}} \arrow[r] & \mathsf{CombSMMC} &
\end{tikzcd}$$

(1) *All of the vertices in the diagram localize to $\mathcal{P}\mathrm{r}\mathcal{SM}$ via the underlying symmetric monoidal ∞-category functor.*
(2) *Assertion (1) remains valid if we replace each vertex by the full subcategory consisting of those objects with a cofibrant unit.*
(3) *Consider the subcategories of each vertex in the diagram spanned by Quillen equivalences. They localize to the maximal sub ∞-groupoid $\mathcal{P}\mathrm{r}\mathcal{SM}^{\simeq} \subset \mathcal{P}\mathrm{r}\mathcal{SM}$ via the underlying symmetric monoidal ∞-category functor.*
(4) *Assertion (3) remains valid if we replace each vertex by the subcategory consisting of those objects with a cofibrant unit and Quillen equivalences between them.*

Remark 6.3. Informally, part (3) of Theorem 6.2 asserts the following:

- presentably symmetric monoidal ∞-categories are presented by combinatorial symmetric monoidal model categories;
- such presentations are unique up to zig-zags of Quillen equivalences;

- such zig-zags are themselves unique; and so on.

More precisely, for every $\mathcal{C} \in \mathcal{P}\mathrm{r}\mathcal{SM}$, the ∞-category

$$\mathsf{CombSMMC}^{QE} \times_{\mathcal{P}\mathrm{r}\mathcal{SM}^{\simeq}} (\mathcal{P}\mathrm{r}\mathcal{SM}^{\simeq})_{/\mathcal{C}}$$

is weakly contractible, where $\mathsf{CombSMMC}^{QE} \subset \mathsf{CombSMMC}$ denotes the subcategory of Quillen equivalences. This follows from the fact that a functor into an ∞-groupoid is a localization if and only if it is final [Cis19a, Proposition 7.1.10].

Distressingly, some of our method does not extend to the monoidal setting. Because of this, the monoidal version of Theorem 6.2 takes the following form:

Theorem 6.4. *Consider the diagram of categories introduced in Section 1:*

$$\begin{tikzcd}
 & \mathsf{TractMMC}_{\mathbf{S},\mathrm{semi}} \arrow[dr] & \\
\mathsf{TractMMC}_{\mathbf{S}} \arrow[ur] \arrow[rr] \arrow[d] & & \mathsf{TractMMC}_{\mathrm{semi}} \\
\mathsf{CombMMC}_{\mathbf{S}}. & &
\end{tikzcd}$$

(1) All of the vertices in the diagram localize to $\mathcal{P}\mathrm{r}\mathcal{M}\mathsf{on}$ via the underlying monoidal ∞-category functor.
(2) Assertion (1) remains valid if we replace each vertex by the full subcategory consisting of those objects with a cofibrant unit.
(3) Consider the subcategories of each vertex in the diagram spanned by Quillen equivalences. They localize to the maximal sub ∞-groupoid $\mathcal{P}\mathrm{r}\mathcal{M}\mathsf{on}^{\simeq} \subset \mathcal{P}\mathrm{r}\mathcal{M}\mathsf{on}$ via the underlying monoidal ∞-category functor.
(4) Assertion (3) remains valid if we replace each vertex by the subcategory consisting of those objects with a cofibrant unit and Quillen equivalences between them.

Remark 6.5. Part (3) of Theorem 6.4 implies that every tractable monoidal semi-model is Quillen equivalent to a simplicial one. This gives a partial solution to an open problem by Hovey [Hov, Problem 10], which asks whether every monoidal model category is Quillen equivalent to a simplicial one.

We now turn to the proof of these theorems. A key step in the proof is a functorial replacement of semi-monoidal model categories by simplicial ones. The basic idea is that of Dugger [Dug01b], who observed that if $\mathbf{M}$ is nice model category, the category $s\mathbf{M}$ of simplicial objects can be made into a simplicial model category by taking the Bousfield localization of the Reedy model structure. In the situation at hand, his idea does not apply directly, so we need a bit more work.

We will start by recalling Dugger's idea. Most of the following lemma can be found in [Dug01b]. We present a slightly more streamlined proof than loc. cit.

Lemma 6.6. *Let $\mathbf{M}$ be a tractable semi-model category. The Reedy semi-model category $s\mathbf{M}_{\mathrm{Reedy}}$ of simplicial objects in $\mathbf{M}$ admits a tractable left Bousfield localization, denoted by $s\mathbf{M}_{\mathrm{Reedy,loc}}$, with the following properties:*

(1) Fibrant objects of $s\mathbf{M}_{\mathrm{Reedy,loc}}$ are the Reedy fibrant simplicial objects X such that the map $X_0 \to X_n$ is a weak equivalence for all $n \geq 0$.
(2) The diagonal functor $\delta\colon \mathbf{M} \to s\mathbf{M}_{\mathrm{Reedy,loc}}$ is a right Quillen equivalence.
(3) $s\mathbf{M}_{\mathrm{Reedy,loc}}$ is a sSet-enriched semi-model category.

Proof. Consider the functor

$$\left(\mathbf{M}^{\mathbf{\Delta}^{\mathrm{op}}}\right)_{\infty} \simeq (\mathbf{M}_{\infty})^{\mathbf{\Delta}^{\mathrm{op}}} \xrightarrow{\mathrm{colim}} \mathbf{M}_{\infty},$$

where the first equivalence is that of [Cis19a, Theorem 7.9.8]. Since $\mathbf{\Delta}^{\mathrm{op}}$ is weakly contractible, the right adjoint of this functor is fully faithful [Lur09, Corollary 4.4.4.10]. Thus, applying Proposition A.19 to this functor, we obtain a tractable semi-model structure $s\mathbf{M}_{\mathrm{Reedy,loc}}$ on $s\mathbf{M}$ satisfying conditions (1) and (2).

For (3), we must describe the simplicial enrichment of $s\mathbf{M}$. If X and Y are simplicial objects in $s\mathbf{M}$, we define the mapping simplicial object $\mathrm{Map}(X,Y)$ as follows: Given a simplicial set S, we define a new simplicial object $S \otimes X \in s\mathbf{M}$ by $(S \otimes X)_n = \coprod_{S_n} X_n$. We then define $\mathrm{Map}(X,Y)_n = \mathrm{Hom}_{s\mathbf{M}}(\Delta^n \otimes X, Y)$. This makes $s\mathbf{M}$ into a simplicial category.

We must show that with this enrichment, $s\mathbf{M}_{\mathrm{Reedy,loc}}$ is an sSet-enriched model category. Let $(-)^S \colon s\mathbf{M} \to s\mathbf{M}$ denote the right adjoint of the functor $S \otimes -$. We must prove the following:

(i) If $i \colon K \to L$ is a cofibration in sSet and $f \colon X \to Y$ is a Reedy cofibration of Reedy cofibrant objects of $s\mathbf{M}$, then the induced map
$$i \square f \colon (K \otimes Y) \amalg_{K \otimes X} (L \otimes X) \to L \otimes Y$$
is a Reedy cofibration.

(ii) In the situation of (i), if further f is a weak equivalence of $s\mathbf{M}_{\mathrm{Reedy,loc}}$, then so is $i \square f$.

(iii) In the situation of (i), if further i is anodyne, then $i \square f$ is a weak equivalence of $s\mathbf{M}_{\mathrm{Reedy,loc}}$.

To prove these assertions, choose generating sets I and J of cofibrations and trivial cofibrations of $\mathbf{M}$ whose elements have a cofibrant domain. Also, for each $n \geq 0$, write $i_n \colon \partial\Delta^n \hookrightarrow \Delta^n$ for the inclusion. The cofibrations trivial cofibrations of Reedy semi-model structure on $s\mathbf{M}$ is generated by the sets $\widetilde{I} = \{i_n \square e \mid n \geq 0,\, e \in I\}$ and $\widetilde{J} = \{i_n \square e \mid n \geq 0,\, e \in J\}$, which are maps of cofibrant objects (see [Bar10, Lemma 3.34]). Here we identified maps in $\mathbf{M}$ with those in $s\mathbf{M}$ via the diagonal functor $\mathbf{M} \to s\mathbf{M}$. In order to prove (i), we may assume that f is an element of $\widetilde{I}$, say $f = i_n \square e$. In this case, the claim is clear, because $i \square i_n$ is a cofibration simplicial sets and the maps $\{i_m\}_{m \geq 0}$ generate cofibrations of simplicial sets.

For (ii), by two out of three and part (i), it suffices to show that the maps $K \otimes f$ and $L \otimes f$ are weak equivalences. We will show that $K \otimes f$ is one; the proof for $L \otimes f$ is similar. By part (i) and A.4, this will follow if we show that $(-)^K \colon s\mathbf{M}_{\mathrm{Reedy,loc}} \to s\mathbf{M}_{\mathrm{Reedy,loc}}$ preserves fibrations of fibrant objects. The explicit generating set $\widetilde{J}$ makes it clear that $(-)^K$ carries Reedy fibrations to Reedy fibrations. Since a map of fibrant objects of $s\mathbf{M}_{\mathrm{Reedy,loc}}$ is a fibration if and only if it is a Reedy fibration (Proposition A.21), we are therefore reduced to showing that $(-)^K$ preserves fibrant objects of $s\mathbf{M}_{\mathrm{Reedy,loc}}$.

Let $X \in s\mathbf{M}_{\mathrm{Reedy,loc}}$ be fibrant. We already know that X^K is Reedy fibrant, so it suffices to show that the map $\theta \colon \left(X^K\right)_0 \to \left(X^K\right)_n$ is a weak equivalence for all n. For this, set $\hom(S,X) = \lim_{\Delta^n \to S \in \left(\mathbf{\Delta}_{/S}\right)^{\mathrm{op}}} X_n$ for each simplicial set S. We can then identify θ with the map $\hom\big(K \times \Delta^0, X\big) \to \hom(K \times \Delta^n, X)$. To prove that this is a weak equivalence, it suffices to show that the map
$$\hom\big(\Delta^0, X\big) \to \hom(\Delta^m, X)$$
is a weak equivalence (see the argument of [Cis19a, Proposition 4.6.1]) for every $m \geq 0$. This follows from our assumption on X.

Finally, for (iii), by part (i) and A.4, it suffices to show that for each fibration $p \colon E \to B$ of fibrant objects of $s\mathbf{M}_{\mathrm{Reedy,loc}}$, the map
$$i \pitchfork p \colon E^L \to E^K \times_{B^K} B^L$$

is a trivial fibration. We already know from (ii) that this is a fibration, so it suffices to show that the map $E^L \to E^K$ and $B^L \to B^K$ are weak equivalences. Using (the argument of) [Cis19a, Proposition 4.6.1], it will suffice to prove this in the case where i is a map of the form $\Delta^0 \to \Delta^n$. In this case, the claim is proved as in the previous paragraph, using the fact that E and B are fibrant in $s\mathbf{M}_{\mathrm{Reedy,loc}}$. □

Unfortunately, when $\mathbf{M}$ is a monoidal semi-model category, it seems difficult to compare $s\mathbf{M}_{\mathrm{Reedy,loc}}$ with $\mathbf{M}$ via a monoidal left Quillen functor. The problem is that the diagonal functor $\mathbf{M} \to s\mathbf{M}$ is rarely left Quillen. Thus, what we are going to do musicis to increase the class of cofibrations to force this to be true.

To state the next lemma, it will be useful to introduce the following terminology.

Definition 6.7. We say that a semi-model category $\mathbf{M}$ is **κ-tractable** if it satisfies the following pair of conditions:

(1) The underlying category of $\mathbf{M}$ is locally κ-presentable.
(2) The (essentially small) class of cofibrations of κ-compact cofibrant objects of $\mathbf{M}$ generates the class of cofibrations of $\mathbf{M}$.

Lemma 6.8. *Let κ be an uncountable regular cardinal, and let $\mathbf{M}$ be a κ-tractable semi-model category. There is a κ-tractable semi-model structure on $s\mathbf{M}$, denoted by $s\mathbf{M}_{\kappa\text{-inj,loc}}$ with the following properties:*

(1) Cofibrations are generated by the maps $X \to Y$ such that, for each $n \geq 0$, the map $X_n \to Y_n$ is a cofibration of κ-compact cofibrant objects of $\mathbf{M}$.
(2) Weak equivalences are the hocolim-equivalenes, i.e., maps inducing weak equivalences between the homotopy colimits.
(3) The diagonal functor $\delta\colon \mathbf{M} \to s\mathbf{M}_{\mathrm{inj,loc}}$ is a left Quillen equivalence.
(4) $s\mathbf{M}_{\mathrm{inj,loc}}$ is a sSet-enriched semi-model category.
(5) If $\mathbf{M}$ is a monoidal semi-model category, then $s\mathbf{M}_{\mathrm{inj,loc}}$ is also one for the degreewise tensor product.

Proof. We first construct a κ-tractable semi-model structure satisfying conditions (1) through (3), using Theorem A.17. (Note that, since κ is uncountable, κ-compact objects of $s\mathbf{M}$ are simply the levelwise κ-compact simplicial objects [Low13, Proposition 2.23].) Call a map $X \to Y$ of $s\mathbf{M}$ an **κ-injective cofibration** if it is a cofibration in the sense of condition (1). We also say that a map of $s\mathbf{M}$ is a **κ-injective fibration** if it has the right lifting property for κ-injective cofibrations. We must verify the following:

(i) The full subcategory of $\mathrm{Fun}([1], s\mathbf{M})$ spanned by the hocolim-equivalences is an accessible subcategory.
(ii) Every κ-injective fibration is a hocolim-equivalence.
(iii) Given a pushout diagram

$$\begin{tikzcd} X \arrow[r] \arrow[d, "i"'] & X' \arrow[d, "i'"] \\ Y \arrow[r] & Y' \end{tikzcd}$$

of κ-injectively cofibrant objects, if i is a hocolim-equivalence and a κ-injective cofibration, so is i'.
(iv) Given a limit ordinal α and a transfinite sequence $X_\bullet\colon \alpha \to s\mathbf{M}$ of κ-injective cofibrations of κ-injectively cofibrant objects, if the map $X_\beta \to X_{\beta+1}$ is a hocolim-equivalence for each $\beta < \alpha$, then so is the map $X_0 \to \mathrm{colim}_{\beta<\alpha} X_\beta$.

For part (i), by [Lur09, Proposition 5.4.6.6], it suffices to show that the composite

$$\mathbf{M}^{\mathbf{\Delta}^{\mathrm{op}}} \to \left(\mathbf{M}^{\mathbf{\Delta}^{\mathrm{op}}}\right)_\infty \simeq (\mathbf{M}_\infty)^{\mathbf{\Delta}^{\mathrm{op}}} \xrightarrow{\mathrm{colim}} \mathbf{M}_\infty$$

is accessible. The first functor is accessible by Corollary A.16, and the second functor is accessible because it is a left adjoint. Thus, their composite is accessible.

For part (ii), let $p\colon X \to Y$ be a κ-injective fibration of $s\mathbf{M}$. We claim that p is levelwise a weak equivalence (hence a hocolim-equivalence). Since $\mathbf{M}$ is κ-tractable, it suffices to show that every lifting problem as depicted in the left-hand diagram below admits a solution, provided that i is a cofibration of κ-compact objects of $\mathbf{M}$:

$$\begin{tikzcd} A \arrow[r] \arrow[d, "i"'] & X_n \arrow[d, "p"] \\ B \arrow[r] \arrow[ur, dashed] & Y_n \end{tikzcd} \qquad \begin{tikzcd} \Delta^n \otimes \delta(A) \arrow[r] \arrow[d, "\Delta^n \otimes \delta(i)"'] & X \arrow[d, "p"] \\ \Delta^n \otimes \delta(B) \arrow[r] \arrow[ur, dashed] & Y. \end{tikzcd}$$

The lifting problem is equivalent to the one on the right, which is solvable because $\Delta^n \otimes \delta(i)$ is a κ-injective cofibration.

Par (iii) is clear, because such squares are homotopy cocartesian in $\mathbf{M}^{\mathbf{\Delta}^{\mathrm{op}}}$ (as it is levelwise homotopy cocartesian). Part (iv) is proved similarly.

Next, we prove (4). Let $i\colon K \to L$ be a monomorphism of simplicial sets, and let $f\colon X \to Y$ be a generating κ-injective cofibration $s\mathbf{M}_{\kappa\text{-inj,loc}}$. We must show that the map

$$i\square f\colon (K \otimes Y) \amalg_{K\otimes X} (L \otimes X) \to L \otimes Y$$

is a κ-injective cofibration, and that it is a weak equivalence if either i or f is one. For the claim on cofibration, it suffices to prove this in the case where i is the boundary inclusion $\partial\Delta^n \subset \Delta^n$ for some $n \geq 0$. In this case, $i\square f$ is another generating κ-injective cofibration, so there is nothing to prove. For the acyclic part of the claim, find a commutative diagram

$$\begin{tikzcd} X' \arrow[r, "f'"] \arrow[d, "p"'] & Y' \arrow[d, "q"] \\ X \arrow[r, "f"] & Y \end{tikzcd}$$

where p and q are trivial Reedy fibrations, X' is Reedy cofibrant, and f' is a Reedy cofibration. We then have a diagram

$$\begin{tikzcd} (K \otimes Y') \amalg_{K\otimes X'} (L \otimes X') \arrow[r] \arrow[d] & L \otimes Y' \arrow[d] \\ (K \otimes Y) \amalg_{K\otimes X} (L \otimes X) \arrow[r] & L \otimes Y \end{tikzcd}$$

whose vertical arrows are levelwise weak equivalences (because the pushouts are levelwise homotopy pushouts). Thus, it suffices to show that $i\square f'$ is a weak equivalence. This follows from Lemma 6.6.

Finally, we prove (5). Suppose that $\mathbf{M}$ is a monoidal model category. We must show that $s\mathbf{M}_{\kappa\text{-inj,loc}}$ is also a monoidal model category. The only nontrivial part is the acyclic part of the pushout-product axiom. For this, it suffices to show that for each cofibrant object $X \in s\mathbf{M}_{\kappa\text{-inj,loc}}$, the functors $X \otimes -, - \otimes X\colon s\mathbf{M}_{\kappa\text{-inj,loc}} \to s\mathbf{M}_{\kappa\text{-inj,loc}}$ preserve weak equivalences of cofibrant objects. This follows from the fact that $\mathbf{\Delta}^{\mathrm{op}}$ is a sifted ∞-category [Lur09, Lemma 5.5.8.4], because this guarantees that $\operatorname{hocolim}_{\mathbf{\Delta}^{\mathrm{op}}}(X \otimes Y) \simeq (\operatorname{hocolim}_{\mathbf{\Delta}^{\mathrm{op}}} X) \otimes (\operatorname{hocolim}_{\mathbf{\Delta}^{\mathrm{op}}} Y)$ whenever $X, Y \in s\mathbf{M}$ are levelwise cofibrant. The proof is now complete. $\square$

Proof of Theorems 6.2 and 6.4. We consider the symmetric monoidal case; the monoidal case is similar. We start with Part (1). We treat each case separately:

- CombSMMC: This is the content of Theorem 2.4.

- $\mathsf{TractSMMC}$: The proof of Theorem 2.4 goes through by replacing $\mathsf{CombSMMC}$ by $\mathsf{TractSMMC}$.
- $\mathsf{TractSMMC}_{\mathbf{S}}$ and $\mathsf{CombSMMC}_{\mathbf{S}}$ and $\mathsf{TractSMMC}_{\mathbf{S},\mathrm{semi}}$: The proof of Proposition 2.6 goes through by replacing $\mathsf{CombSym}\mathbf{S}\text{-}\mathsf{Alg}$ by these categories. (In the semi-model categorical case, we replace various model-categorical results by the results in Section A. Also, in this case, we only need $\mathbf{S}$ to be a left proper tractable symmetric monoidal semi-model category.) Thus, the claim for these categories follow from Proposition 2.5.
- $\mathsf{TractSMMC}_{\mathrm{semi}}$: By the previous point, it suffices to show that the forgetful functor
$$\mathsf{TractSMMC}_{\mathsf{sSet},\mathrm{semi}} \to \mathsf{TractSMMC}_{\mathrm{semi}}$$
induces an equivalence upon localizing at Quillen equivalences.

  For each uncountable regular cardinal κ, we let $\mathsf{TractSMMC}_{\mathrm{semi}}(\kappa) \subset \mathsf{TractSMMC}_{\mathrm{semi}}$ denote the subcategory spanned by the objects whose underlying semi-model categories are κ-tractable, and the morphisms that preserve κ-compact objects. We define $\mathsf{TractSMMC}_{\mathsf{sSet},\mathrm{semi}}(\kappa)$ similarly. Lemma 6.8 gives us a functor
$$\Phi\colon \int^{\mathsf{urCard}} \mathsf{TractSMMC}_{\mathrm{semi}}(-) \to \mathsf{TractSMMC}_{\mathsf{sSet},\mathrm{semi}}(\kappa),$$
$$(\kappa, \mathbf{M}) \mapsto s\mathbf{M}_{\kappa\text{-inj},\mathrm{loc}},$$
which fits into the (non-commutative) diagram
$$\begin{tikzcd}
\int^{\mathsf{urCard}} \mathsf{TractSMMC}_{\mathsf{sSet},\mathrm{semi}}(-) \arrow[r, "U"] \arrow[d, "\pi"'] & \int^{\mathsf{urCard}} \mathsf{TractSMMC}_{\mathrm{semi}}(-) \arrow[d, "\pi'"] \arrow[dl, "\Phi"'] \\
\mathsf{TractSMMC}_{\mathsf{sSet},\mathrm{semi}} \arrow[r, "U'"'] & \mathsf{TractSMMC}_{\mathrm{semi}}.
\end{tikzcd}$$
Here π, π', U, U' are all forgetful functors. Since $\mathsf{TractSMMC}_{\mathsf{sSet},\mathrm{semi}}$ is the colimit of $\mathsf{TractSMMC}_{\mathsf{sSet},\mathrm{semi}}(\kappa)$ as κ ranges over urCard, the functor π and π' are localizations [Lur25, Tag 02UU]. Thus, it suffices to show that the diagram commutes up to equivalence when we localize these categories by Quillen equivalences.

By Lemma 6.8, the lower triangle commutes up to natural Quillen equivalence. For the upper triangle, recall from the previous bullet point that the composite
$$\mathsf{TractSMMC}_{\mathsf{sSet},\mathrm{semi}} \xrightarrow{U'} \mathsf{TractSMMC}_{\mathrm{semi}} \to \mathcal{PrSM}$$
is a localization at Quillen equivalences. We also have a natural Quillen equivalence $U'\pi \xrightarrow{\simeq} U'\Phi U$ by Lemma 6.8, so this proves the claim on the lower triangle.

Part (2) follows from part (1) and Muro's theorem [Mur15, Theorem 1] or its enriched version (Proposition D.1). Parts (3) and (4) are proved similarly, noting that the proof of Theorem 2.4 is unaffected if we restrict our attention to the relevant subcategories. This completes the proof of the theorem. □

Appendix A. Review of semi-model categories

Semi-model categories are a generalization of model categories, first introduced by Hovey and Spitzweck [Hov98, Spi01] and later developed further by Barwick [Bar10]. (See [BW24] for a historical account and examples of semi-model categories.) For the most part, they behave very much like (and sometimes better than) model categories. There are, however, some differences, and some results about model-categories will be false without making some modifications to their statements. In this section, we review the definition of semi-model categories and

give precise statements of results on semi-model categories that directly generalize the model-categorical counterparts.

A.1. **Definition.** In this subsection, we recall the definition of semi-model categories.

Definition A.1. Let $\mathbf{M}$ be a locally presentable category, and let I be a set of morphisms of $\mathbf{M}$. An **I-cofibration** is a retract of a transfinite composition of pushouts of elements of I. An **I-injective** is a morphisms having the right lifting property for the maps in I. We write I-cof and I-inj for the classes of I-cofibrations and I-injectives, respectively.

Definition A.2. [BW24, Definition 2.1] A **semi-model structure** on a category $\mathbf{M}$ consists of classes (W, F, C) of weak equivalences, fibrations, and cofibrations, satisfying the following axioms:

- M1 Fibrations are closed under pullback.
- M2 Weak equivalences have the two out of three property.
- M3 The classes W, F, C contain isomorphisms and is closed under composition and retracts.
- M4
  - i Cofibrations have the left lifting property for trivial fibrations (i.e., elements of $W \cap F$).
  - ii Trivial cofibrations with cofibrant domains have the left lifting property for fibrations. (Here, trivial cofibrations refer to elements in $C \cap W$, and cofibrant objects are those objects such that the map from the initial object is a cofibration.)
- M5
  - i Every morphism in $\mathbf{M}$ can be functorially factored as a cofibration followed by a trivial fibration.
  - ii Every morphism whose domain is cofibrant can be functorially factored as a trivial cofibration followed by a fibration.

A **semi-model category** is a finitely bicomplete[5] category $\mathbf{M}$ equipped with a semi-model structure. Its **underlying ∞-category** is the (∞-categorical) localization at weak equivalences.

We say that a semi-model category $\mathbf{M}$ is **combinatorial** if it satisfies the following pair of conditions:

- $\mathbf{M}$ is locally presentable as a category.
- There are (small) sets I, J of morphisms of $\mathbf{M}$, such that the classes of fibrations and trivial fibrations are exactly the classes of J-injectives and I-injectives. Elements of the sets I and J are called **generating cofibrations** and **generating trivial cofibrations**.

If further the maps in I and J have cofibrant domains, then we say that $\mathbf{M}$ is **tractable**.

Remark A.3. Let $\mathbf{M}$ be a semi-model category, and let f be a morphism of $\mathbf{M}$. By the retract argument [Hov99, Lemma 1.1.9], one has the following:

- If f is a map of cofibrant objects and has the left lifting property for fibrations, then it is a trivial cofibration.
- f is a cofibration if and only if it has the left lifting property for trivial fibrations.
- f is a trivial fibration if and only if it has the right lifting property for cofibrations.

This implies the following:

[5] In [BW24], bicompleteness is assumed, but we drop this to have a semi-model categorical version of the arguments in 4.2.

- A semi-model structure is completely determined by the class of fibrations and weak equivalences.
- In a tractable model category $\mathbf{M}$ with generating sets I, J of cofibrations and trivial cofibrations with cofibrant domains, the maps in I are cofibrations, and the maps in J are trivial cofibrations.

Remark A.4. Let $\mathbf{M}$ be a semi-model category. The argument of [JT07, Lemma 7.14] shows that a cofibration of cofibrant objects of $\mathbf{M}$ is a trivial cofibration if and only if it has the left lifting property for fibrations of fibrant objects.

Definition A.5. A functor $F\colon \mathbf{M} \to \mathbf{N}$ of tractable semi-model categories is called a **left Quillen functor** $\mathbf{M} \to \mathbf{N}$ if it satisfies the following conditions:

(1) F is a left adjoint.
(2) F carries cofibrations of cofibrant objects to cofibrations.
(3) F carries trivial cofibrations of cofibrant objects to trivial cofibrations.

Equivalently, F is left Quillen if its right adjoint preserves fibrations and trivial fibrations.

Remark A.6. Let $\mathbf{M}$ be a semi-model category. By [Cis19a, Proposition 7.7.4], the underlying ∞-category of $\mathbf{M}$ has small colimits. If $\mathbf{M}$ is tractable, then the semi-model categorical version of Proposition 4.12 shows that the underlying ∞-category of $\mathbf{M}$ is presentable.

A.2. **Reedy semi-model structure.** Given a model category $\mathbf{M}$ and a Reedy category $\mathcal{C}$, the functor category $\mathrm{Fun}(\mathcal{C}, \mathbf{M})$ admits a model structure, called the Reedy model structure. In this subsection, we will see that this directly generalizes to the semi-model categorical setting (Proposition A.7). As a consequence of this, we will obtain a formula for the mapping space of the underlying ∞-category of a semi-model category using (co)simplicial resolutions (Corollary A.9).

Proposition A.7 (Reedy semi-model structure)**.** *Let $\mathbf{M}$ be a bicomplete semi-model category, and let $\mathcal{C}$ be a Reedy category. There is a semi-model structure on* $\mathrm{Fun}(\mathcal{C}, \mathbf{M})$*, called the* ***Reedy semi-model structure****, with the following properties:*

(1) Weak equivalences are the natural weak equivalences.
(2) A map $\alpha\colon F \to G$ is a fibration (resp. trivial fibration) if and only if, for each $C \in \mathcal{C}$, the map

$$FC \to GC \times_{M_C G} M_C F$$

is a fibration (resp. trivial fibration) of $\mathbf{M}$. Here, $M_C F$ denotes the matching object of F at C.
(3) A map $\alpha\colon F \to G$ is a cofibration if and only if, for each $C \in \mathcal{C}$, the map

$$L_C G \amalg_{L_C F} FC \to GC$$

is a cofibration of $\mathbf{M}$. Here, $L_C F$ denotes the latching object of F at C.
(4) Given a cofibrant $F \in \mathbf{M}^{\mathcal{J}}$, a map $\alpha\colon F \to G$ is a trivial cofibration if and only if, for each $C \in \mathcal{C}$, the map

$$L_C G \amalg_{L_C F} FC \to GC$$

is a trivial cofibration of $\mathbf{M}$.

Moreover, if $\mathbf{M}$ is tractable, so is the Reedy semi-model structure on $\mathrm{Fun}(\mathcal{C}, \mathbf{M})$.

Proof. See the proof of [Hir03, Theorems 15.3.4 and 15.3.15] or [Hov99, Theorem 5.2.5] for the construction of the semi-model structure. For tractability, see [Bar10, Lemma 3.34]. □

We are particularly interested in the Reedy semi-model structure for (co)simplicial objects, as it can be used to model derived mapping spaces (Corollary A.9.) If $\mathbf{M}$ is a tractable semi-model category, a **simplicial resolution** in $\mathbf{M}$ is a Reedy fibrant simplicial object $X_\bullet$ such that the map $X_0 \to X_n$ is a weak equivalence for every $n \geq 0$. Dually, a cosimplicial resolution in $\mathbf{M}$ is a Reedy cofibrant cosimplicial object $X^\bullet$ such that the map $X^n \to X^0$ is a weak equivalence for every $n \geq 0$. We then have the following proposition:

Proposition A.8. *Let $\mathbf{M}$ be a semi-model category.*

(1) Let $i\colon A \to B$ be a cofibration of cofibrant objects of $\mathbf{M}$, and let $p\colon X_\bullet \to Y_\bullet$ be a Reedy fibration of simplicial resolutions of $\mathbf{M}$. The map

$$\mathbf{M}(B, X_\bullet) \to \mathbf{M}(B, Y_\bullet) \times_{\mathbf{M}(A, Y_\bullet)} \mathbf{M}(A, X_\bullet)$$

is a Kan fibration, which is a trivial fibration if i or p is a weak equivalence.

(2) Let $i\colon A^\bullet \to B^\bullet$ be a Reedy cofibration of cosimplicial resolutions in $\mathbf{M}$, and let $p\colon X \to Y$ be a fibration of $\mathbf{M}$. The map

$$\mathbf{M}(B^\bullet, X) \to \mathbf{M}(B^\bullet, Y) \times_{\mathbf{M}(A^\bullet, Y)} \mathbf{M}(A^\bullet, X)$$

is a Kan fibration, which is a trivial fibration if i or p is one.

Proof. Identical to [Hir03, Theorem 16.5.2]. □

Corollary A.9. *Let $\mathbf{M}$ be a semi-model category, let $L\colon \mathbf{M} \to \mathcal{M}$ be the localization at weak equivalences, and let $X_\bullet$ be a simplicial resolution in $\mathbf{M}$. There is a natural equivalence*

$$\mathbf{M}(A, X_\bullet) \simeq \mathcal{M}(L(A), L(X_0))$$

of functors $\mathbf{M}^{\mathrm{op}}_{\mathrm{cof}} \to \mathsf{sSet}[\mathrm{weq}^{-1}] \simeq \mathcal{S}$, which carries a vertex $A \to X_0 \in \mathbf{M}(A, X_\bullet)$ to its image in $\mathcal{M}$.

Proof. Identical to [ACK25, Corollary 3.1]. □

Corollary A.10. *Let $\mathbf{M}$ be a semi-model category. The weak equivalences of $\mathbf{M}$ are exactly the maps whose images in the underlying ∞-category $\mathcal{M}$ are equivalences. Moreover, if two objects in $\mathbf{M}$ have equivalent images in $\mathcal{M}$, then they are connected by a zig-zag of weak equivalences.*

Proof. This is immediate from A.9. □

Corollary A.11. *Let $\mathbf{M}$ be a semi-model category, and let $\mathcal{C}$ be a small category. Postcomposing with the functor $\mathbf{M} \to \mathbf{M}[\mathrm{weq}^{-1}]$ induces an equivalence of ∞-categories*

$$\mathrm{Fun}(\mathcal{C}, \mathbf{M})[\mathrm{weq}^{-1}] \xrightarrow{\simeq} \mathrm{Fun}\big(\mathcal{C}, \mathbf{M}[\mathrm{weq}^{-1}]\big).$$

Proof. This follows from [Cis19a, Theorem 7.9.8] and Corollary A.10. □

Corollary A.9 has another important consequence:

Proposition A.12. *Let $\mathbf{M}$ be a semi-model category, and let $L\colon \mathbf{M} \to \mathcal{M}$ be the localization at weak equivalences. Consider an extension problem in $\mathbf{M}$ depicted as*

$$\begin{tikzcd} A \arrow[r, "f"] \arrow[d, "i"'] & X. \\ B \arrow[ur, dashed] & \end{tikzcd}$$

Suppose that:

(1) i is a cofibration of cofibrant objects.
(2) X is fibrant.
(3) There is a morphism $g\colon B \to X$ and a homotopy $L(g) \circ L(i) \simeq L(f)$ in $\mathcal{M}$.

Then there is a map $g' \colon B \to X$ such that $g'i = f$ in $\mathbf{M}$ and a homotopy $L(g) \simeq L(g')$ in $\mathcal{M}$.

Proof. A closer inspection of the proof of Proposition A.7 shows that the constant cosimplicial object $\delta(A)$ at A admits a trivial Reedy fibration $A^\bullet \to \delta(A)$ from a cosimplicial resolution, such that its 0th degree is the identity map of A. By the same token, the composite $A^\bullet \to \delta(A) \to \delta(B)$ can be factored as

$$\begin{tikzcd} A^\bullet \arrow[r] \arrow[d, "i'"'] & \delta(A) \arrow[d, "\delta(i)"] \\ B^\bullet \arrow[r, "p"'] & \delta(B) \end{tikzcd}$$

where i' is a Reedy cofibration and p is a Reedy fibration whose 0th degree is the identity map of B. By the dual of Corollary A.9, we can think of a homotopy $L(g)L(i) \simeq L(f)$ as a path $\alpha \colon gi \to f$ in the Kan complex $\mathbf{M}(A^\bullet, X)$. By Proposition A.8, the map

$$\mathbf{M}(B^\bullet, X) \to \mathbf{M}(A^\bullet, X)$$

is a Kan fibration, so the path α can be lifted to a morphism $g \to g'$ in $\mathbf{M}(B^\bullet, X)$. The map g' has the desired property. □

A.3. **Homotopy colimits in Semi-model categories.** Let $\mathbf{M}$ be a semi-model category, let $\mathcal{C}$ be a small category, and let $F \colon \mathcal{C}^{\triangleright} \to \mathbf{M}$ be a diagram. We say that F is a **homotopy colimit diagram** if its image in the underlying ∞-category of $\mathbf{M}$ is a colimit diagram. In this subsection, we prove several results to detect homotopy colimit diagrams (Proposition A.15 and A.16).

We start with the following result on cofibrantly generating semi-model structures:

Proposition A.13. *Let $\mathbf{M}$ be a locally presentable category, and let I and J be sets of maps of $\mathbf{M}$. Call I-cofibrations cofibrations, and suppose that the domains of maps in I and J are cofibrant (i.e., the map from the initial object is a cofibration). Let W be a class of maps of that is closed under retracts and satisfies the two out of three property. Suppose that:*

(1) $I\text{-inj} = J\text{-inj} \cap W$.
(2) $(J\text{-cof})_c \subset I\text{-cof} \cap W$, *where the subscript c indicates the subclass of maps with cofibrant domains.*

Then $\mathbf{M}$ becomes a tractable semi-model category with generating sets I and J and with weak equivalences W.

Proof. Declare a map to be a fibration if it is a J-injective. We claim that this, together with the cofibrations and weak equivalences given, determines a semi-model structure on $\mathbf{M}$. Axioms M1 through M3 hold trivially. Axiom M4-i holds by (1), and axiom M4-ii holds by definition. Axiom M5-i and M5-ii holds by (1) and (2), respectively, using the small object argument. □

Corollary A.14 (Projective semi-model structure)**.** *Let $\mathbf{M}$ be a tractable semi-model category, and let $\mathcal{C}$ be a small category. The functor category* $\mathrm{Fun}(\mathcal{C}, \mathbf{M})$ *admits a tractable semi-model structure, called the* ***projective semi-model structure****, whose weak equivalences and fibrations are the natural transformations whose components are weak equivalences and fibrations, respectively.*

Proof. Let I, J be generating sets of cofibrations and trivial cofibrations of $\mathbf{M}$ with cofibrant domains. Replacing I by $I \cup J$, we may assume that $J \subset I$. Consider the following sets of maps of $\mathrm{Fun}(\mathcal{C}, \mathbf{M})$:

$$I_{\mathcal{C}} = \{\mathcal{C}(C, -) \cdot i \mid C \in \mathcal{C},\, i \in I\},\ J_{\mathcal{C}} = \{\mathcal{C}(C, -) \cdot j \mid C \in \mathcal{C}, j \in J\}.$$

Note that $I_{\mathcal{C}}$-injectives are exactly the natural transformations whose components are trivial fibrations, and $J_{\mathcal{C}}$-injectives are the natural transformations whose components are fibrations. We claim that the sets $I_{\mathcal{C}}$ and $J_{\mathcal{C}}$ and the class of natural weak equivalences satisfy conditions (1) and (2) of Proposition A.13.

Conditions (1) is clear from the characterization of $I_{\mathcal{C}}$-injectives and $J_{\mathcal{C}}$-injectives we gave in the previous paragraph. For (2), observe that the components of a $J_{\mathcal{C}}$-cofibration with cofibrant domain are trivial cofibrations. It follows that $J_{\mathcal{C}}$-cofibrations with cofibrant domain are natural weak equivalences. Since $J_{\mathcal{C}} \subset I_{\mathcal{C}}$, every $J_{\mathcal{C}}$-cofibration is also an $I_{\mathcal{C}}$-cofibration. Hence condition (2) is satisfied, and the proof is complete. □

Corollary A.14 allows us to detect homotopy colimit diagrams in semi-model categories.

Proposition A.15. *Let* $\mathbf{M}$ *be a tractable semi-model category or a combinatorial model category, let* $\mathcal{C}$ *be a small category, and let* $F\colon \mathcal{C}^{\triangleright} \to \mathbf{M}$ *be a colimit diagram. If* $F|\mathcal{C}$ *is projectively cofibrant, then* F *is a homotopy colimit diagram.*

Proof. Let $i\colon \mathcal{C} \hookrightarrow \mathcal{C}^{\triangleright}$ denote the inclusion, and consider the adjunction

$$i_!\colon \operatorname{Fun}(\mathcal{C}, \mathbf{M}) \underset{\longleftarrow}{\overset{\longrightarrow}{\perp}} \operatorname{Fun}(\mathcal{C}^{\triangleright}, \mathbf{M})\colon i^*$$

between the left Kan extension functor $i_!$ and the restriction functor i^*. By the theory of derived adjunctions [Rie14, Theorem 2.2.11], this adjunction induces an adjunction

$$\mathbb{L}i_!\colon \operatorname{ho}(\operatorname{Fun}(\mathcal{C}, \mathbf{M})) \underset{\longleftarrow}{\overset{\longrightarrow}{\perp}} \operatorname{ho}(\operatorname{Fun}(\mathcal{C}^{\triangleright}, \mathbf{M}))\colon \mathbb{R}i^*$$

of homotopy categories. Explicitly, $\mathbb{R}i^* = \operatorname{ho}(i^*)$ and $\mathbb{L}i_! = \operatorname{ho}(i_! \circ Q)$, where Q is a projective cofibrant replacement functor of $\operatorname{Fun}(\mathcal{C}, \mathbf{M})$. (Note that $i_! \circ Q$ preserves weak equivalences by Ken Brown's lemma [Cis19a, Corollary 7.4.14].)

Now using Corollary A.11, we can rewrite the above adjunction as

$$\mathbb{L}i_!\colon \operatorname{ho}(\operatorname{Fun}(\mathcal{C}, \mathcal{M})) \underset{\longleftarrow}{\overset{\longrightarrow}{\perp}} \operatorname{ho}(\operatorname{Fun}(\mathcal{C}^{\triangleright}, \mathcal{M}))\colon \operatorname{ho}(i^*),$$

where $\mathcal{M} = \mathbf{M}[\mathrm{weq}^{-1}]$ denotes the underlying ∞-category of $\mathbf{M}$. By the uniqueness of left adjoints, it follows that $\mathbb{L}i_! \cong \operatorname{ho}(i_!)$, where $i_!\colon \operatorname{Fun}(\mathcal{C}, \mathcal{M}) \to \operatorname{Fun}(\mathcal{C}^{\triangleright}, \mathcal{M})$ denotes the left Kan extension functor. (It exists by Remark A.6.) Thus, the image of F in $\operatorname{Fun}(\mathcal{C}^{\triangleright}, \mathcal{M})$ lies in the essential image of the left Kan extension functor $i_!$, meaning that it is a colimit diagram in $\mathcal{M}$. □

Corollary A.16. *Let* $\mathbf{M}$ *be a tractable semi-model category or a combinatorial model category, let* κ *be a regular cardinal. Suppose that* $\mathbf{M}$ *admits a generating set of cofibrations consisting of maps of* κ*-compact objects. Then:*

(1) Weak equivalences of $\mathbf{M}$ *are stable under* κ*-filtered colimits.*
(2) The localization functor $L\colon \mathbf{M} \to \mathbf{M}[\mathrm{weq}^{-1}]$ *preserves* κ*-filtered colimits.*

Proof. For both (1) and (2), it suffices to consider diagrams indexed by κ-filtered categories, because every κ-filtered ∞-category admits a final functor from a κ-filtered category [Lur25, Tag 02QA].

Part (2) follows from part (1) and Proposition A.15, so it suffices to prove (1). For this, let $\mathcal{C}$ be a κ-filtered category, and let $\alpha\colon F \xrightarrow{\simeq} G$ be a natural weak equivalence of $\operatorname{Fun}(\mathcal{C}, \mathbf{M})$. We wish to show that $\operatorname{colim}_{\mathcal{C}} \alpha$ is a weak equivalence.

For this, find a commutative diagram

$$\begin{tikzcd} F' \arrow[r, "\alpha'"] \arrow[d] & G' \arrow[d] \\ F \arrow[r, "\alpha"] & G \end{tikzcd}$$

where the vertical arrows are projective trivial fibrations, and F' and G' are projectively cofibrant. Since the functor $\operatorname{colim}_{\mathcal{C}}$ is left Quillen for the projective (semi-)model structure, Ken Brown's lemma [Cis19a, Corollary 7.4.14] shows that it preserves weak equivalences of projectively cofibrant objects. Thus $\operatorname{colim}_{\mathcal{C}} \alpha'$ is a weak equivalence. Hence, by two out of three property of weak equivalences, it suffices to show that $\operatorname{colim}_{\mathcal{C}}$ carries projective trivial fibrations to trivial fibrations. This follows from our assumption on $\mathbf{M}$. □

A.4. **Jeff Smith's theorem and Bousfield localization.** In a tractable semi-model category, fibrations are exactly the maps having the right lifting property for trivial cofibrations of cofibrant objects. Therefore, a tractable semi-model structure is completely determined by the class of cofibrations and weak equivalences. This prompts the following question: Given a locally presentable category and a class of cofibrations and weak equivalences, when does it determine a tractable semi-model structure?

Surprisingly, there is a complete answer to this question (Theorem A.17). Using this, we will show that semi-model categories enjoy excellent properties for Bousfield localization (Proposition A.19).

We start by giving a precise answer to the question above, summarized in the following theorem. In the model-categorical setting, the theorem is often attributed to Jeff Smith.

Theorem A.17. *Let $\mathbf{M}$ be a locally presentable category equipped with two classes (C, W) of maps, whose elements are called cofibrations and weak equivalences. Then (C, W) is part of a tractable semi-model structure on $\mathbf{M}$ if and only if the following conditions are satisfied:*

1. *The class W, considered as a full subcategory of $\operatorname{Fun}([1], \mathbf{M})$, is accessible and accessibly embedded.*
2. *The class W is closed under retracts and has the two out of three property.*
3. *If a map has the right lifting property for cofibrations, then it is a weak equivalence.*
4. *There is a small set I of maps of cofibrant objects such that $C = I\text{-cof}$.*
5. *Given a pushout diagram*

$$\begin{tikzcd} A \arrow[r] \arrow[d, "i"'] & A' \arrow[d, "i'"] \\ B \arrow[r] & B' \end{tikzcd}$$

   in $\mathbf{M}$, if A and A' cofibrant (i.e., maps from the initial objects are cofibrations) and $i \in C \cap W$, then $i' \in C \cap W$.
6. *The maps in $C \cap W$ with cofibrant domains are closed under transfinite composition.*

Proof. Sufficiency is proved in [BW24, Theorem B].[6] For necessity, suppose that (C, W) is part of a tractable semi-model structure on $\mathbf{M}$. We must check that

[6] In [BW24], it is required that weak equivalences with cofibrant domains span an accessible and accessibly embedded full subcategory. However, the proof goes through without cofibrancy assumption, as verified via private communication with David White.

conditions (1) through (6) are satisfied. Conditions (2), (3), (6) hold trivially, and conditions (4) and (5) hold by Remark A.4. For part (1), we recall that from Corollary A.16 that the functor

$$\mathbf{M} \to \mathbf{M}[\mathrm{weq}^{-1}]$$

is accessible. Since the target of this functor is presentable (Remark A.6), the accessibility of W follows from [Lur09, Proposition 5.4.6.6] and Corollary A.10. □

We can use Theorem A.17 to construct Bousfield localizations of semi-model categories.

Definition A.18. Let $\mathbf{M}$ be a semi-model category. A **left Bousfield localization** of $\mathbf{M}$ is a semi-model category $\mathbf{M}'$ having the same underlying category and cofibrations as $\mathbf{M}$, but which has more weak equivalences than $\mathbf{M}$.

Proposition A.19. *Let $\mathbf{M}$ be a tractable semi-model category with underlying ∞-category $\mathcal{M}$. Let $F\colon \mathcal{M} \to \mathcal{N}$ be a functor admitting a fully faithful right adjoint G, where $\mathcal{N}$ is presentable. There is a tractable left Bousfield localization $\mathbf{N}$ of $\mathbf{M}$ with the following properties:*

(a) The weak equivalences are the maps inverted by the composite $\mathbf{M} \to \mathcal{M} \xrightarrow{F} \mathcal{N}$.
(b) The induced functor $\mathbf{N} \to \mathcal{N}$ is a localization at weak equivalences.
(c) The fibrant objects of $\mathbf{N}$ are the fibrant objects of $\mathbf{M}$ whose image in $\mathcal{M}$ lies in the essential image of G.

Proof. We first check that there is a tractable left Bousfield localization of $\mathbf{M}$ satisfying condition (a). For this, we will use Theorem A.17 to the class C of cofibrations of $\mathbf{M}$ and the class W of maps inverted by the composite $\mathbf{M} \to \mathcal{M} \xrightarrow{F} \mathcal{N}$. We must check conditions (1) through (6) of loc. cit. Condition (1) follows from Corollary A.16 and [Lur09, Proposition 5.4.6.6]. Conditions (2), (3), (4) hold trivially. Condition (5) and (6) follow from Proposition A.15, because the relevant diagrams are homotopy colimit diagrams. (Note that the projective semi-model structures for spans and transfinite sequences agree with the Reedy semi-model structures. Thus projectively cofibrant diagrams are easy to identify in these cases.)

We next check that the semi-model category $\mathbf{N}$ we just constructed satisfies conditions (b) and (c).

For condition (b), notice that F is a localization at the maps it inverts [Lur25, Tag 04JL]. It follows that the composite $\mathbf{M} \to \mathcal{M} \xrightarrow{F} \mathcal{N}$ is a localization at W, as claimed.

Finally, for (c), note that [Rie14, Theorem 2.2.11], the adjunction $\mathrm{id}\colon \mathbf{M} \underset{\longleftarrow}{\overset{\longrightarrow}{\perp}} \mathbf{N}\colon \mathrm{id}$ gives rise to the derived adjunction

$$\mathbb{L}\,\mathrm{id}\colon \mathrm{ho}(\mathcal{M}) \underset{\longleftarrow}{\overset{\longrightarrow}{\perp}} \mathrm{ho}(\mathcal{N})\colon \mathbb{R}\,\mathrm{id}\,.$$

Explicitly, $\mathbb{L}\,\mathrm{id} = \mathrm{ho}(F)$, and $\mathbb{R}\,\mathrm{id} = \mathrm{ho}(R)$, where R is a fibrant replacement functor of $\mathbf{N}$. By the uniqueness of right adjoints, $\mathbb{R}\,\mathrm{id}$ is isomorphic to $\mathrm{ho}(G)$. This implies the following:

- If $X \in \mathbf{M}$ is fibrant in $\mathbf{N}$, then its image in $\mathcal{M}$ lies in the essential image of G.
- If $X \in \mathbf{M}$ is an object whose image in $\mathcal{M}$ lies in the essential image of G, then X is connected by a zig-zag of weak equivalences of $\mathbf{M}$ to a fibrant object of $\mathbf{N}$ (by Corollary A.10).

The claim then follows from Proposition A.12. □

A model-categorical version of Proposition A.19 is well-known to the experts, but we could not find a reference. We record it here for future reference:

Proposition A.20. *Let* $\mathbf{M}$ *be a combinatorial left proper model category with underlying* ∞*-category* $\mathcal{M}$*, and let* $F\colon \mathcal{M} \to \mathcal{N}$ *be a functor admitting a fully faithful right adjoint* G*, where* $\mathcal{N}$ *is presentable. There is a left Bousfield localization* $\mathbf{N}$ *of* $\mathbf{M}$ *(in the sense of model categories) with the following properties:*

(a) The weak equivalences are the maps inverted by the composite $\mathbf{M} \to \mathcal{M} \xrightarrow{F} \mathcal{N}$*.*
(b) The induced functor $\mathbf{N} \to \mathcal{N}$ *is a localization at weak equivalences.*
(c) The fibrant objects of $\mathbf{N}$ *are the fibrant objects of* $\mathbf{M}$ *whose image in* $\mathcal{M}$ *lies in the essential image of* G*.*

Proof. Identical to Proposition A.19, using [Bar10, Proposition 2.2] instead of Theorem A.17. (To check (6), we use [Hir03, Proposition 17.9.3]). □

We conclude this subsection with the following recognition result for fibrations in Bousfield localization:

Proposition A.21. *Let* $\mathbf{M}$ *be a tractable semi-model category, let* $L\mathbf{M}$ *be a tractable left Bousfield localization of* $\mathbf{M}$*, and let* $p\colon X \to Y$ *be a map of fibrant objects of* $L\mathbf{M}$*. Then* p *is a fibration of* $L\mathbf{M}$ *if and only if it is a fibration of* $\mathbf{M}$*.*

Proof. In a tractable semi-model category, fibrations are exactly the maps having the right lifting property for the trivial cofibrations of cofibrant objects. Thus, the "only if" part is clear. For the "if" part, suppose that p is a fibration of $\mathbf{M}$. Consider a lifting problem

$$\begin{tikzcd} A \arrow[r] \arrow[d, "i"'] & X \arrow[d, "p"] \\ B \arrow[r] \arrow[ur, dashed] & Y \end{tikzcd}$$

where i is a trivial cofibration of $L\mathbf{M}$ and A is cofibrant. We must show that there is a dashed filler rendering the diagram commutative. Factor i as $A \xrightarrow{j} A' \xrightarrow{q} B$, where j is a trivial cofibration of $\mathbf{M}$ and q is (necessarily) a weak equivalence of $L\mathbf{M}$. Since p is a fibration of $\mathbf{M}$, there is a dashed filler depicted as

$$\begin{tikzcd} A \arrow[r, equal] \arrow[d, "j"'] & A \arrow[r] \arrow[d, "i"'] & X \arrow[d, "p"] \\ A' \arrow[r] \arrow[urr, dashed] & B \arrow[r] & Y \end{tikzcd}$$

Applying Proposition A.12 to the semi-model category $(L\mathbf{M})_{/Y}$, we find that the original lifting problem also admits a solution. The proof is now complete. □

Appendix B. Multiplicative Gabriel–Ulmer duality

The Gabriel–Ulmer duality [GU71] is one precise formulation of the slogan that "locally presentable categories are controlled by small data." In the ∞-categorical setting, one formulation of this is that for every regular cardinal κ, the Ind_κ-completion functor induces an equivalence of ∞-categories

$$\mathrm{Ind}_\kappa\colon \mathcal{C}\mathsf{at}_\infty^{\mathrm{idem}}(\kappa) \xrightarrow{\simeq} \mathcal{P}\mathsf{r}^L(\kappa).$$

Here $\mathcal{C}\mathsf{at}_\infty^{\mathrm{idem}}(\kappa)$ denotes the ∞-category of small idempotent complete ∞-categories with κ-small colimits and functors preserving κ-small colimits, and $\mathcal{P}\mathsf{r}^L(\kappa)$ denotes the ∞-category of κ-presentable ∞-categories[7] and functors that preserve κ-compact objects and small colimits.

[7] An ∞-category is **κ-presentable** if it is κ-accessible and presentable.

The goal of this section is to enhance this equivalence to that of symmetric monoidal ∞-categories (Theorem B.8) when κ is uncountable. As a corollary of this, we obtain a Gabriel–Ulmer duality for κ-presentably symmetric monoidal ∞-categories (Corollary B.12) for an uncountable κ.

We start by reviewing symmetric monoidal structures on $\mathcal{P}\mathsf{r}^L$ and $\mathcal{C}\mathsf{at}_\infty$.

Recollection B.1. [Lur17, Corollary 4.8.1.4] Let $\mathcal{K}$ be a small set of small simplicial sets. We write $\mathcal{C}\mathsf{at}_\infty(\mathcal{K}) \subset \mathcal{C}\mathsf{at}_\infty$ for the subcategory whose objects are the ∞-categories admitting colimits indexed by the elements in $\mathcal{K}$, whose morphisms are the functors that preserve such colimits.

The ∞-category $\mathcal{C}\mathsf{at}_\infty(\mathcal{K})$ can be enhanced to a symmetric monoidal ∞-category, which we denote by $\mathcal{C}\mathsf{at}_\infty(\mathcal{K})^\otimes$. An active map $(\mathcal{A}_1, \dots, \mathcal{A}_k) \to \mathcal{B}$ is given by a functor $\mathcal{A}_1 \times \cdots \times \mathcal{A}_k \to \mathcal{B}$ preserving colimits indexed by the elements of $\mathcal{K}$ in each variable. Such a map is cocartesian if and only if, for any other $\mathcal{C} \in \mathcal{C}\mathsf{at}_\infty(\kappa)$, the functor

$$\mathrm{Fun}^{\mathcal{K}}(\mathcal{B}, \mathcal{D}) \to \mathrm{Fun}^{\mathcal{K}}(\mathcal{A}_1 \times \cdots \times \mathcal{A}_k, \mathcal{C})$$

is an equivalence. Here, $\mathrm{Fun}^{\mathcal{K}}$ denotes the ∞-category of functors preserving colimits indexed by the elements in $\mathcal{K}$ in each variable.

In the case where $\mathcal{K}$ is the set of (representatives of isomorphism classes of) all κ-small simplicial sets, where κ is a regular cardinal, we write $\mathcal{C}\mathsf{at}_\infty(\mathcal{K}) = \mathcal{C}\mathsf{at}_\infty(\kappa)$ and $\mathrm{Fun}^{\mathcal{K}} = \mathrm{Fun}^\kappa$.

Recollection B.2. [Lur17, Proposition 4.8.1.15] Applying Recollection B.1 to the very large ∞-category $\widehat{\mathcal{C}\mathsf{at}}_\infty$ of large ∞-categories and the collection $\mathcal{U}$ of all small simplicial sets, we obtain the symmetric monoidal ∞-category $\widehat{\mathcal{C}\mathsf{at}}_\infty(\mathcal{U})^\otimes$ of large ∞-categories with small colimits, and functors preserving such colimits. The full subcategory $\mathcal{P}\mathsf{r}^L \subset \widehat{\mathcal{C}\mathsf{at}}_\infty(\mathcal{U})$ is stable under tensor products, so it inherits a symmetric monoidal structure. We denote the resulting symmetric monoidal ∞-category by $\left(\mathcal{P}\mathsf{r}^L\right)^\otimes$.

We can relate $\mathcal{C}\mathsf{at}_\infty(\kappa)^\otimes$ with $\left(\mathcal{P}\mathsf{r}^L\right)^\otimes$ by Ind_κ-completion (Corollary B.5). For this, we recall the following propositions:

Proposition B.3. [Lur17, Remark 4.8.1.8] *Let $\mathcal{K} \subset \mathcal{L}$ be an inclusion of small sets of small simplicial sets. The inclusion*

$$\mathcal{C}\mathsf{at}_\infty(\mathcal{L})^\otimes \subset \mathcal{C}\mathsf{at}_\infty(\mathcal{K})^\otimes$$

admits a left adjoint relative to Fin_* (in the sense of [Lur17, Definition 7.3.2.2]*), and the left adjoint $\mathcal{C}\mathsf{at}_\infty(\mathcal{K})^\otimes \to \mathcal{C}\mathsf{at}_\infty(\mathcal{L})^\otimes$ is symmetric monoidal.*

Proposition B.4. [Lur09, Propositions 5.3.5.10 and 5.5.1.9] *Let $\mathcal{A}$ be a small ∞-category with κ-small colimits. The functor $\mathcal{A} \to \mathrm{Ind}_\kappa(\mathcal{A})$ has the following universal property: For every ∞-category $\mathcal{Z}$ with small colimits, the functor*

$$\mathrm{Fun}^L(\mathrm{Ind}_\kappa(\mathcal{A}), \mathcal{Z}) \to \mathrm{Fun}^\kappa(\mathcal{A}, \mathcal{Z})$$

is an equivalence, where the left-hand side denotes the ∞-category of functors preserving small colimits.

Corollary B.5. *Let κ be a regular cardinal. The assignment $\mathcal{A} \mapsto \mathrm{Ind}_\kappa(\mathcal{A})$ can be enhanced to a symmetric monoidal functor*

$$\mathrm{Ind}_\kappa \colon \mathcal{C}\mathsf{at}_\infty(\kappa)^\otimes \to \left(\mathcal{P}\mathsf{r}^L\right)^\otimes.$$

Proof. In light of Proposition B.4, this is a special case of Corollary B.5. □

We now use the functor of Corollary B.5 to obtain the classical Gabriel–Ulmer duality. To state it, write $\overline{\mathcal{C}\mathsf{at}_\infty}(\kappa) \subset \widehat{\mathcal{C}\mathsf{at}}_\infty$ for the subcategory whose objects are the essentially small ∞-categories with κ-small colimits, and whose morphisms are the functors that preserve κ-small colimits. We also write $\mathcal{C}\mathsf{at}_\infty^{\mathrm{idem}}(\kappa) \subset \mathcal{C}\mathsf{at}_\infty(\kappa)$ and $\overline{\mathcal{C}\mathsf{at}_\infty^{\mathrm{idem}}}(\kappa) \subset \overline{\mathcal{C}\mathsf{at}_\infty}(\kappa)$ for the full subcategories of idempotent complete objects.

Corollary B.6. *Let κ be a regular cardinal. The functor of Corollary B.5 induces an equivalence of ∞-categories*

$$\mathrm{Ind}_\kappa \colon \mathcal{C}\mathsf{at}_\infty^{\mathrm{idem}}(\kappa) \xrightarrow{\simeq} \mathcal{P}\mathsf{r}^L(\kappa).$$

The inverse equivalence is given by the forgetful functor

$$(-)_\kappa \colon \mathcal{P}\mathsf{r}^L(\kappa) \to \overline{\mathcal{C}\mathsf{at}_\infty^{\mathrm{idem}}}(\kappa) \simeq \mathcal{C}\mathsf{at}_\infty^{\mathrm{idem}}(\kappa).$$

Proof. By construction, the functor $\mathrm{Ind}_\kappa \colon \mathcal{C}\mathsf{at}_\infty^{\mathrm{idem}}(\kappa) \to \mathcal{P}\mathsf{r}^L$ is equipped with a natural transformation $\{\eta_{\mathcal{A}} \colon \mathcal{A} \to \mathrm{Ind}_\kappa(\mathcal{A})\}_{\mathcal{A} \in \mathcal{C}\mathsf{at}_\infty^{\mathrm{idem}}(\kappa)}$ of functors $\mathcal{C}\mathsf{at}_\infty(\kappa) \to \widehat{\mathcal{C}\mathsf{at}}_\infty$, where each $\eta_{\mathcal{A}}$ exhibits $\mathrm{Ind}_\kappa(\mathcal{A})$ as an Ind_κ-completion of $\mathcal{A}$. According to [Lur09, Lemma 5.4.2.4], each $\eta_{\mathcal{A}}$ restricts to an equivalence $\mathcal{A} \xrightarrow{\simeq} \mathrm{Ind}_\kappa(\mathcal{A})_\kappa$. Therefore, the functor $\mathrm{Ind}_\kappa \colon \mathcal{C}\mathsf{at}_\infty^{\mathrm{idem}}(\kappa) \to \mathcal{P}\mathsf{r}^L$ takes values in $\mathcal{P}\mathsf{r}^L(\kappa)$.

To show that the resulting functor $\mathrm{Ind}_\kappa \colon \mathcal{C}\mathsf{at}_\infty^{\mathrm{idem}}(\kappa) \to \mathcal{P}\mathsf{r}^L(\kappa)$ is an equivalence, we first observe that the functor $(-)_\kappa \colon \mathcal{P}\mathsf{r}^L(\kappa) \to \overline{\mathcal{C}\mathsf{at}_\infty^{\mathrm{idem}}}(\kappa) \simeq \mathcal{C}\mathsf{at}_\infty^{\mathrm{idem}}(\kappa)$ is a right adjoint of Ind_κ with unit η. Indeed, for each $\mathcal{A} \in \mathcal{C}\mathsf{at}_\infty^{\mathrm{idem}}(\kappa)$ and $\mathcal{X} \in \mathcal{P}\mathsf{r}^L(\kappa)$, Proposition B.4 and the aforementioned equivalence $\mathcal{A} \xrightarrow{\simeq} \mathrm{Ind}_\kappa(\mathcal{A})_\kappa$ gives an equivalence

$$\mathrm{Fun}_{\mathcal{P}\mathsf{r}^L(\kappa)}(\mathrm{Ind}_\kappa(\mathcal{A}), \mathcal{X}) \xrightarrow{\simeq} \mathrm{Fun}_\kappa(\mathcal{A}, \mathcal{X}_\kappa),$$

where the left-hand side is the full subcategory of $\mathrm{Fun}(\mathrm{Ind}_\kappa(\mathcal{A}), \mathcal{X})$ spanned by the functors in $\mathcal{P}\mathsf{r}^L(\kappa)$. Now for each $\mathcal{X} \in \mathcal{P}\mathsf{r}^L(\kappa)$, the counit $\mathrm{Ind}_\kappa(\mathcal{X}_\kappa) \to \mathcal{X}$ is also an equivalence by [Lur09, Propositions 5.3.5.11 and 5.4.2.2.]. Therefore, Ind_κ and $(-)_\kappa$ is part of an adjoint equivalence. □

We now upgrade Corollary B.6 to a symmetric monoidal equivalence (Theorem B.8) and look at its consequences.

Definition B.7. Let κ be a regular cardinal. We write $\mathcal{P}\mathsf{r}^L(\kappa)^\otimes \subset \left(\mathcal{P}\mathsf{r}^L\right)^\otimes$ for the suboperad spanned by the κ-presentable ∞-categories and those maps $(\mathcal{C}_1, \dots, \mathcal{C}_n) \to \mathcal{C}$ whose corresponding functors $F \colon \mathcal{C}_1 \times \cdots \times \mathcal{C}_n \to \mathcal{C}$ have the following property:

- Given an object $(X_1, \dots, X_n) \in \mathcal{C}_1 \times \cdots \times \mathcal{C}_n$, if each X_i is κ-presentable, then so is $F(X_1, \dots, X_n)$.

Theorem B.8. *Let κ be a regular cardinal. Then:*

(1) $\mathcal{P}\mathsf{r}^L(\kappa)^\otimes$ is a symmetric monoidal ∞-category, and the inclusion $\mathcal{P}\mathsf{r}^L(\kappa)^\otimes \hookrightarrow \left(\mathcal{P}\mathsf{r}^L\right)^\otimes$ is symmetric monoidal.

(2) The symmetric monoidal functor $\mathrm{Ind}_\kappa \colon \mathcal{C}\mathsf{at}_\infty(\kappa)^\otimes \to \left(\mathcal{P}\mathsf{r}^L\right)^\otimes$ of Corollary B.5 restricts to a symmetric monoidal functor

$$\mathrm{Ind}_\kappa \colon \mathcal{C}\mathsf{at}_\infty(\kappa)^\otimes \to \left(\mathcal{P}\mathsf{r}^L(\kappa)\right)^\otimes,$$

which is an equivalence if κ is uncountable.

Proof. Part (2) follows from part (1) and Corollary B.6, so we only have to prove (1). For this, it suffices to prove the following:

(1′) Let $\mathcal{X}, \mathcal{Y}$ be κ-presentable ∞-categories. Then:

(1′-a) Their tensor product $\mathcal{X} \otimes_{\mathcal{P}\mathsf{r}^L} \mathcal{Y}$ in $\mathcal{P}\mathsf{r}^L$ is κ-presentable.

(1′-b) Every object of $(\mathcal{X} \otimes_{\mathcal{P}\mathsf{r}^L} \mathcal{Y})_\kappa$ is a retract of an object lying in the image of the functor $\mathcal{X}_\kappa \times \mathcal{Y}_\kappa \to (\mathcal{X} \otimes_{\mathcal{P}\mathsf{r}^L} \mathcal{Y})_\kappa$.

To prove (1′-a), use Corollary B.6 to find objects $\mathcal{A}, \mathcal{B} \in \mathcal{C}\mathsf{at}_\infty(\kappa)$ and equivalences $\mathrm{Ind}_\kappa(\mathcal{A}) \xrightarrow{\simeq} \mathcal{X}$ and $\mathrm{Ind}_\kappa(\mathcal{B}) \xrightarrow{\simeq} \mathcal{Y}$. For each $\mathcal{Z} \in \mathcal{P}\mathsf{r}^L$, Proposition B.4 gives an equivalence

$$\begin{aligned}\mathrm{Fun}^L(\mathcal{X} \otimes_{\mathcal{P}\mathsf{r}^L} \mathcal{Y}, \mathcal{Z}) &\simeq \mathrm{Fun}^L\big(\mathcal{X}, \mathrm{Fun}^L(\mathcal{Y}, \mathcal{Z})\big)\\ &\simeq \mathrm{Fun}^\kappa(\mathcal{A}, \mathrm{Fun}^\kappa(\mathcal{B}, \mathcal{Z}))\\ &\simeq \mathrm{Fun}^\kappa\big(\mathcal{A} \otimes_{\mathcal{C}\mathsf{at}_\infty(\kappa)} \mathcal{B}, \mathcal{Z}\big)\\ &\simeq \mathrm{Fun}^L\big(\mathrm{Ind}_\kappa\big(\mathcal{A} \otimes_{\mathcal{C}\mathsf{at}_\infty(\kappa)} \mathcal{B}\big), \mathcal{Z}\big).\end{aligned}$$

It follows that $\mathcal{X} \otimes_{\mathcal{P}\mathsf{r}^L} \mathcal{Y}$ is equivalent to $\mathrm{Ind}_\kappa\big(\mathcal{A} \otimes_{\mathcal{C}\mathsf{at}_\infty(\kappa)} \mathcal{B}\big)$. Since the latter is is κ-presentable, this proves (1′-a).

For part (1′-b), note that we can identify the map $\mathcal{X}_\kappa \times \mathcal{Y}_\kappa \to (\mathcal{X} \otimes_{\mathcal{P}\mathsf{r}^L} \mathcal{Y})_\kappa$ with the composite

$$\mathcal{A} \times \mathcal{B} \to \mathcal{A} \otimes_{\mathcal{C}\mathsf{at}_\infty(\kappa)} \mathcal{B} \to \mathrm{Ind}_\kappa\big(\mathcal{A} \otimes_{\mathcal{C}\mathsf{at}_\infty(\kappa)} \mathcal{B}\big)_\kappa.$$

The first functor is essentially surjective (see the proof of [Lur17, Proposition 4.8.1.3]). Also, by [Lur09, Lemma 5.4.2.4] the second functor identifies its target as an idempotent completion of the source. Hence (1′-b) follows. □

Remark B.9. We do not know if $\mathcal{C}\mathsf{at}_\infty^{\mathrm{idem}}(\omega)$ is closed under tensor products in $\mathcal{C}\mathsf{at}_\infty(\omega)$. This is why we imposed the uncountability assumption in part (2) of Theorem B.8.

We conclude this section with three corollaries of Theorem B.8. For the following corollary, we make $\overline{\mathcal{C}\mathsf{at}}_\infty(\kappa)$ into a symmetric monoidal ∞-category $\overline{\mathcal{C}\mathsf{at}}_\infty(\kappa)^\otimes$ using Recollection B.1.

Corollary B.10. *Let κ be an uncountable regular cardinal. The forgetful functor*

$$(-)_\kappa \colon \mathcal{P}\mathsf{r}^L(\kappa)^\otimes \to \overline{\mathcal{C}\mathsf{at}}_\infty(\kappa)^\otimes$$

is an equivalence of symmetric monoidal ∞-categories.

Proof. The composite $\mathrm{Ind}_\kappa \colon \mathcal{C}\mathsf{at}_\infty(\kappa)^\otimes \xrightarrow{\mathrm{Ind}_\kappa} \mathcal{P}\mathsf{r}^L(\kappa)^\otimes \xrightarrow{(-)_\kappa} \overline{\mathcal{C}\mathsf{at}}_\infty(\kappa)^\otimes$ is equivalent to the inclusion, which is an equivalence of symmetric monoidal ∞-categories. Since Ind_κ is an equivalence by Theorem B.8, we are done. □

Definition B.11. Let $\mathcal{C}$ be a monoidal ∞-category, and let κ be an uncountable regular cardinal. We say that $\mathcal{C}$ is **κ-presentably monoidal** if it satisfies the following conditions:

(1) The underlying ∞-category of $\mathcal{C}$ is κ-presentable.
(2) For every $n \geq 0$ and every collection of κ-presentable objects $X_1, \dots, X_n \in \mathcal{C}$, the object $X_1 \otimes \cdots \otimes X_n$ is κ-presentable. (In particular, the unit object is κ-presentable).

We write κ-$\mathcal{P}\mathrm{r}\mathcal{SM} \subset \mathcal{SM}\widehat{\mathcal{C}\mathsf{at}}_\infty$ for the subcategory of κ-presentably monoidal ∞-categories and monoidal functors that preserves small colimits and κ-presentable objects.

Corollary B.12. *Let κ be an uncountable regular cardinal. The forgetful functor*

$$\kappa\text{-}\mathcal{P}\mathrm{r}\mathcal{SM} \to \mathcal{SM}\overline{\mathcal{C}\mathsf{at}}(\kappa),\ \mathcal{C}^\otimes \mapsto \mathcal{C}^\otimes_\kappa$$

is a categorical equivalence, where $\mathcal{SM}\overline{\mathcal{C}\mathsf{at}}_\infty(\kappa) \subset \mathcal{SM}\widehat{\mathcal{C}\mathsf{at}}_\infty$ denotes the subcategory of essentially small symmetric monoidal ∞-categories compatible with κ-small colimits, and those monoidal functors preserving κ-small colimits.

Proof. This follows from Corollary B.10 and the straightening–unstraightening equivalence. □

Corollary B.13. *Let κ be an uncountable regular cardinal, and let $I\colon \mathcal{SMC}\mathsf{at}_\infty(\kappa) \to \kappa\text{-}\mathcal{PrSM}$ be a functor, denoted by $\mathcal{C}^\otimes \mapsto I(\mathcal{C})^\otimes$. Suppose there is a natural transformation depicted as*

$$\begin{tikzcd} \mathcal{SMC}\mathsf{at}_\infty(\kappa) \arrow[rr, "I"] \arrow[dr, hook] & \arrow[d, phantom, "\overset{\alpha}{\Longrightarrow}"] & \kappa\text{-}\mathcal{PrSM} \arrow[dl, hook] \\ & \widehat{\mathcal{SMC}\mathsf{at}_\infty}, & \end{tikzcd}$$

with the following property: For each $\mathcal{C} \in \mathcal{SMC}\mathsf{at}_\infty(\kappa)$, the map $\alpha_{\mathcal{C}}\colon \mathcal{C} \to I(\mathcal{C})$ exhibits $I(\mathcal{C})$ as an Ind_κ-completion of $\mathcal{C}$. Then I is a categorical equivalence.

Proof. The natural transformation α determines a natural equivalence

$$\mathcal{C}^\otimes \xrightarrow{\simeq} I(\mathcal{C})^\otimes_\kappa$$

from the inclusion $\iota\colon \mathcal{SMC}\mathsf{at}_\infty(\kappa) \hookrightarrow \mathcal{SM}\overline{\mathcal{C}\mathsf{at}}_\infty(\kappa)$ to the composite $\mathcal{SMC}\mathsf{at}_\infty(\kappa) \xrightarrow{I} \kappa\text{-}\mathcal{PrSM} \xrightarrow{(-)_\kappa} \mathcal{SM}\overline{\mathcal{C}\mathsf{at}}_\infty(\kappa)$. Since the functors ι and $(-)_\kappa$ are equivalences (Corollary B.12), so must be I. □

Appendix C. Base change of localization

Let $L\colon \mathcal{C} \to \mathcal{D}$ be a localization of ∞-categories, and suppose we are given a pullback square in $\mathcal{C}\mathsf{at}_\infty$

$$\begin{tikzcd} \mathcal{C}' \arrow[r] \arrow[d, "L'"'] & \mathcal{D} \arrow[d, "L"] \\ \mathcal{D}' \arrow[r] & \mathcal{D}. \end{tikzcd}$$

The functor L' is not a localization in general. This naturally raises a question: When is L' a localization? The goal of this subsection is to record a few results related to this question.

We start with a sufficient condition for a localization functor to be stable under base change; such a functor is called a **universal localization**. (A general criteria for universal localization is discussed in [Hin24].) The following theorem says that localization of model categories, or mo generally, ∞-categories with weak equivalences and fibrations (in the sense of [Cis19a, Definition 7.4.12]) enjoy this property:

Theorem C.1. *Let $\mathcal{C}$ be an ∞-category with weak equivalences and fibrations. The localization functor*

$$\mathcal{C} \to \mathcal{C}[\mathrm{weq}^{-1}]$$

is a universal localization.

The proof of Theorem C.1 requires the following lemma:

Lemma C.2. *Let $\mathcal{C}$ be an ∞-category, let $\mathcal{W} \subset \mathcal{C}$ be a subcategory containing all objects, and let $L\colon \mathcal{C} \to \mathcal{D}$ be a functor of ∞-categories which carries every morphism in $\mathcal{W}$ to an equivalence. If for each $n \geq 0$, the functor*

$$\theta_{L,n}\colon \mathrm{Fun}([n], \mathcal{C}) \times_{\mathcal{C}^{n+1}} \mathcal{W}^{n+1} \to \mathrm{Fun}([n], \mathcal{D})^{\simeq}$$

is a weak homotopy equivalence, then L is a universal localization.

Proof. Let $\mathcal{D}' \to \mathcal{D}$ be a categorical fibration of ∞-categories, and set $\mathcal{C}' = \mathcal{D}' \times_{\mathcal{D}} \mathcal{C}$. We must show that the functor $L' \colon \mathcal{C}' \to \mathcal{D}'$ is a localization. For this, set $\mathcal{W}' = (\mathcal{D}')^{\simeq} \times_{\mathcal{D}} \mathcal{W} \subset \mathcal{C}'$. We claim that L' is a localization at $\mathcal{W}'$. According to the generalized Mazel-Gee's localization theorem [Ara23, Theorem 1.7], it suffices to show that for each $n \geq 0$, the map $\theta_{L',n}$ is a weak homotopy equivalence. But θ'_n is a pullback of the map θ_n along the the Kan fibration $\mathrm{Fun}([n], \mathcal{D}')^{\simeq} \to \mathrm{Fun}([n], \mathcal{D})^{\simeq}$, so the claim follows from the right properness of the Kan–Quillen model structure. (If $\mathcal{W}$ is the subcategory of morphisms mapped to equivalences in $\mathcal{D}$, we can use the original version of Mazel-Gee's theorem [MG19, Theorem 3.8]. See [AC25, Theorem 1.1] for a different proof of this theorem.) □

Proof of Theorem C.1. Let $\mathcal{W} \subset \mathcal{C}$ denote the subcategory of weak equivalences. By [Cis19a, Remark 7.5.22], we may assume that $\mathcal{W}$ is saturated, i.e., that it consists of the morphisms whose images in $L(\mathcal{C}) = \mathcal{C}[\mathrm{weq}^{-1}]$ are equivalences. By Lemma C.2, it will suffice to show that the functor

$$\theta_{\mathcal{C},n} \colon \mathrm{Fun}([n], \mathcal{C}) \times_{\mathcal{C}^{n+1}} \mathcal{W}^{n+1} \to \mathrm{Fun}([n], L(\mathcal{C}))$$

is a weak homotopy equivalence for every $n \geq 0$. By [Cis19a, Theorem 7.4.20], the ∞-category $\mathrm{Fun}([n], \mathcal{C})$ has the structure of an ∞-category with weak equivalences and fibrations whose subcategory of weak equivalences is given by $\mathrm{Fun}([n], \mathcal{C}) \times_{\mathcal{C}^{n+1}} \mathcal{W}^{n+1}$. Moreover, by [Cis19a, Theorem 7.6.17], the functor

$$L(\mathrm{Fun}([n], \mathcal{C})) \to \mathrm{Fun}([n], L(\mathcal{C}))$$

is a categorical equivalence. It follows that $\mathrm{Fun}([n], \mathcal{C})$ is saturated, and the map $\theta_{L,n}$ can be identified with $\theta_{\mathrm{Fun}([n],\mathcal{C}),0}$. Thus, replacing $\mathcal{C}$ by $\mathrm{Fun}([n], \mathcal{C})$, we are reduced to showing that $\theta_{\mathcal{C},0} \colon \mathcal{W} \to L(\mathcal{C})$ is a weak homotopy equivalence. This is the content of [Cis19a, Lemma 7.6.9], and the proof is complete. □

We next consider the interaction of pullback and homotopy equivalences of relative ∞-categories.

Proposition C.3. *Consider a diagram* $[2] \times [1] \to \mathsf{Cat}_\infty$ *depicted as*

$$\begin{tikzcd}
 & & \mathcal{A} \arrow[r, "f"] \arrow[dd, "p"] & \mathcal{B} \arrow[dd, "q"] \\
\mathcal{A}' \arrow[urr] \arrow[r, "f'"] \arrow[dd, "p'"] & \mathcal{B}' \arrow[urr] \arrow[dd, "q'"] & & \\
 & & & \mathcal{C}, \\
\mathcal{C}' \arrow[urrr, "u"] & & &
\end{tikzcd}$$

where the squares are all cartesian. Regard $\mathcal{A}$ and $\mathcal{B}$ (resp. $\mathcal{A}'$ and $\mathcal{B}'$) as relative ∞-categories whose weak equivalences are the maps whose images in $\mathcal{C}$ (resp. $\mathcal{C}'$) are equivalences. Suppose that the following conditions are satisfied:

(1) The functor $f \colon \mathcal{A} \to \mathcal{B}$ is a homotopy equivalence of relative ∞-categories.
(2) The functor $u \colon \mathcal{C}' \to \mathcal{C}$ is conservative (i.e., reflects equivalences).

Then f' is a homotopy equivalence of relative ∞-categories.

Proof. We claim that there are maps $g\colon \mathcal{B} \to \mathcal{A}$ and $i\colon \mathcal{B} \to \mathcal{B}$ in $\mathcal{C}\mathsf{at}_{\infty/\mathcal{C}}$ and a diagram of the form

$$\begin{tikzcd}
\mathcal{B} \times \{0\} \arrow[dr, hook] \arrow[drr, bend left, "fg"] & & \\
& \mathcal{B} \times \mathcal{I} \arrow[r, "\Phi"] & \mathcal{B} \\
\mathcal{B} \times \{1\} \arrow[ur, hook] \arrow[urr, bend right, "i"'] & &
\end{tikzcd} \tag{C.1}$$

in $\mathcal{C}\mathsf{at}_{\infty/\mathcal{C}}$, where:

(I) $\mathcal{I}$ is a weakly contractible ∞-category equipped with two distinguished objects $0, 1 \in \mathcal{I}$.
(II) For each $B \in \mathcal{B}$, the functor $\Phi|\{B\} \times \mathcal{I}$ carries each morphism to a weak equivalence.
(III) The functor i is an equivalence of ∞-categories.
(IV) $\mathcal{B} \times \mathcal{I}$ lies over $\mathcal{C}$ via the composite $\mathcal{B} \times \mathcal{I} \xrightarrow{\mathrm{pr}} \mathcal{B} \xrightarrow{q} \mathcal{C}$.

Pulling back along $u\colon \mathcal{C}' \to \mathcal{C}$, we obtain functors $g'\colon \mathcal{B}' \to \mathcal{A}'$, $i'\colon \mathcal{B}' \to \mathcal{B}'$, and $\Phi'\colon \mathcal{B}' \times \mathcal{I} \to \mathcal{B}'$. Using condition (2), we find that Φ' witnesses the fact that f' has a right homotopy inverse. Applying the same argument to g, we find that g' also has a right homotopy inverse. It then follows from the two out of six property of homotopy equivalences that f' is a homotopy equivalence of relative categories.

To prove the above claim, use condition (1) to find a relative functor $g\colon \mathcal{B} \to \mathcal{A}$ and diagrams of the form (C.1) in $\mathcal{C}\mathsf{at}_\infty$ (not $\mathcal{C}\mathsf{at}_{\infty/\mathcal{C}}$ yet!) that satisfy conditions (I), (II), and (III). Let S denote the set of morphisms of $\mathcal{B} \times \mathcal{I}$ of the form $(B, x) \to (B, y)$, where $B \in \mathcal{B}$ and $x \to y$ is a moprhism in $\mathcal{I}$. By condition (I), $q\Phi$ factors through the localization $(\mathcal{B} \times \mathcal{I})[S^{-1}] \simeq \mathcal{B} \times \mathcal{I}[\mathcal{I}^{-1}]$. Since $\mathcal{I}$ is weakly contractible, this localization can be identified with $\mathcal{B}$, with localizing functor given by the projection $\mathcal{B} \times \mathcal{I} \to \mathcal{B}$. Thus, there is a diagram $[1] \times [1] \to \mathcal{C}\mathsf{at}_\infty$ of the form

$$\begin{tikzcd}
\mathcal{B} \times \mathcal{I} \arrow[r, "\Phi"] \arrow[d, "\mathrm{pr}"'] & \mathcal{B} \arrow[d, "q"] \\
\mathcal{B} \arrow[r, "r"'] & \mathcal{C}
\end{tikzcd}$$

Since $r \simeq (q\Phi)|\mathcal{B} \times \{1\} = q$, we may assume that $r = q$. Then we obtain a diagram $\sigma\colon [2] \to \mathcal{C}\mathsf{at}_\infty$ whose boundary is depicted as

$$\begin{tikzcd}
\mathcal{B} \times \mathcal{I} \arrow[r, "\Phi"] \arrow[dr, "q \circ \mathrm{pr}"'] & \mathcal{B} \arrow[d, "q"] \\
& \mathcal{C}.
\end{tikzcd}$$

We now contemplate the diagrams $([1] \times [1])^{\triangleright} \to \mathcal{C}\mathsf{at}_\infty$ and $[3] \to \mathcal{C}\mathsf{at}_\infty$ depicted as

$$\begin{tikzcd}
\mathcal{B} \times \{0\} \arrow[r, "g"] \arrow[d, hook] \arrow[dr, "fg"] \arrow[ddr, "q"] & \mathcal{A} \arrow[dr, "f"] \arrow[dd, dashed, "p"] & \\
\mathcal{B} \times \mathcal{I} \arrow[r, "\Phi"] \arrow[dr, "q \circ \mathrm{pr}"'] & & \mathcal{B} \arrow[dl, "q"] \\
& \mathcal{C} &
\end{tikzcd}
\qquad
\begin{tikzcd}
& \mathcal{B} \times \mathcal{I} \arrow[r, "\Phi"] \arrow[dr, "q \circ \mathrm{pr}"] & \mathcal{B} \arrow[dl, "q"] \\
\mathcal{B} \times \{1\} \arrow[ur, hook] \arrow[urr, "i"] \arrow[drr, "q"'] & & \\
& & \mathcal{C}
\end{tikzcd}$$

The left-hand diagram is constructed as follows: Think of this as an amalgamation of two 3-simplices $\Delta^3 \to \mathcal{C}\mathsf{at}_\infty$. The front 3-simplex is obtained by filling the horn

$\Lambda^3_1 \subset \Delta^3$, using σ, the natural equivalence $fg \simeq \Phi|\mathcal{B} \times \{0\}$, and the 2-simplex corresponding to the equality $q \circ \mathrm{pr}|\mathcal{B} \times \{0\} = q$. We then fill the 3-simplex in the back by filling the horn $\Lambda^3_2 \subset \Delta^3$. Likewise, the right-hand 3-simplex is obtained by filling the inner horn $\Lambda^3_1 \subset \Delta^3$. These diagrams lifts the maps g, i, and Φ to those in $\mathcal{C}\mathsf{at}_{\infty/\mathcal{C}}$. The lifts give rise to a diagram in $\mathcal{C}\mathsf{at}_{\infty/\mathcal{C}}$ satisfying conditions (I) through (IV). The proof is now complete. □

Appendix D. Enriched Muro's theorem

In this section, we show that every monoidal $\mathbf{V}$-model categories can be made into one with a cofibrant unit by changing the class of cofibrations. The result is a minor generalization of [Mur15, Theorem 1], which establishes the corresponding theorem in the unenriched setting.

Proposition D.1. *Let $\mathbf{V}$ be a combinatorial monoidal model category with a cofibrant unit, and let $\mathbf{M}$ be a combinatorial monoidal $\mathbf{V}$-model category. There is another combinatorial monoidal $\mathbf{V}$-model structure on the underlying $\mathbf{V}$-monoidal category of $\mathbf{M}$, denoted by $\widetilde{\mathbf{M}}$, satisfying the following conditions:*

(1) The weak equivalences of $\widetilde{\mathbf{M}}$ are the weak equivalences of $\mathbf{M}$.
(2) Every cofibration of $\mathbf{M}$ is a cofibration of $\widetilde{\mathbf{M}}$.
(3) The unit object is cofibrant in $\widetilde{\mathbf{M}}$.

Moreover, it satisfies the following universal property:

(4) Let $\mathbf{N}$ be another monoidal $\mathbf{V}$-model category with a cofibrant unit. Then every monoidal left Quillen $\mathbf{V}$-functor $\mathbf{M} \to \mathbf{N}$ extends uniquely to a left Quillen $\mathbf{V}$-functor $\widetilde{\mathbf{M}} \to \mathbf{N}$.

An analogous claim holds for tractable semi-model categories.

Proof. We will follow [Mur15]. We will use the following notation:

- Given a morphism $f\colon A \to A'$ in $\mathbf{V}$ or $\mathbf{M}$ and a morphism $g\colon B \to B'$ in $\mathbf{V}$ or $\mathbf{M}$, we write $f\square g$ for their **pushout product**
$$A \otimes B' \amalg_{A\otimes B} A' \otimes B \to A' \otimes B'$$
whenever the notation makes sense (i.e., either (i) both f and g are maps of $\mathbf{V}$, (ii) f is a morphism of $\mathbf{V}$ and g is a morphism of $\mathbf{M}$, or (iii) both f and g are morphisms of $\mathbf{M}$.)
- If S and T are collections of morphisms in $\mathbf{V}$ or $\mathbf{M}$, we will write $S\square T = \{s\square t \mid s \in S, t \in T\}$ whenever the right-hand side makes sense.
- If $i\colon A \to B$ and $p\colon X \to Y$ are morphisms of $\mathbf{M}$, we will write $\langle i, p\rangle$ for their **pullback power**
$$\mathbf{M}(B, X) \to \mathbf{M}(A, X) \times_{\mathbf{M}(A,Y)} \mathbf{M}(B, Y),$$
where we wrote $\mathbf{M}(-,-)$ for the enriched hom-objects of $\mathbf{M}$.
- The symbol $\mathbf{1}$ will always mean the unit of $\mathbf{M}$; when we need to refer to the unit of $\mathbf{V}$ (which happens only once in the proof), we denote it by $\mathbf{1}_{\mathbf{V}}$.

Choose a weak equivalence $q\colon \widetilde{\mathbf{1}} \to \mathbf{1}$ that witnesses Muro's unit axiom in $\mathbf{M}$. Choose a factorization
$$\widetilde{\mathbf{1}} \amalg \mathbf{1} \xrightarrow{\iota} C \xrightarrow{\pi} \mathbf{1}$$
of the map (q, id), where ι is a cofibration in $\mathbf{M}$ and π is a weak equivalence in $\mathbf{M}$. Let $\mathrm{inc}_1\colon \widetilde{\mathbf{1}} \to \widetilde{\mathbf{1}} \amalg \mathbf{1}$ denote the inclusion of the first summand, and set $\kappa = \iota \circ \mathrm{inc}_1\colon \widetilde{\mathbf{1}} \to C$. Also, choose generating sets $I_{\mathbf{M}}$ and $J_{\mathbf{M}}$ of cofibrations and trivial cofibrations of $\mathbf{M}$, and choose generating sets $I_{\mathbf{V}}$ and $J_{\mathbf{V}}$ of cofibrations and trivial cofibrations of $\mathbf{V}$. By enlarging $I_{\mathbf{V}}$ if necessary, we may assume that the

map $\emptyset \to \mathbf{1}_{\mathbf{V}}$ belongs to $I_{\mathbf{V}}$. Let $\phi\colon \emptyset \to \mathbf{1}$ denote the unique map from the initial object. We set

$$I_{\widetilde{\mathbf{M}}} = I_{\mathbf{M}} \cup (I_{\mathbf{V}} \square \{\phi\}),$$
$$J_{\widetilde{\mathbf{M}}} = J_{\mathbf{M}} \cup (I_{\mathbf{V}} \square \{\kappa\}) \cup (J_{\mathbf{V}} \square \{\phi\}).$$

We claim that there is a cofibrantly generated model structure on $\mathbf{M}$ with generating sets of cofibrations and trivial cofibrations given by $I_{\widetilde{\mathbf{M}}}$ and $J_{\widetilde{\mathbf{M}}}$, and with weak equivalences given by those of $\mathbf{M}$. Write $I_{\widetilde{\mathbf{M}}}$-cof and $I_{\widetilde{\mathbf{M}}}$-inj for the classes of $I_{\widetilde{\mathbf{M}}}$-cofibrations and $I_{\widetilde{\mathbf{M}}}$-injectives; we also make use of similar notations, such as $J_{\mathbf{V}}$-cof and $J_{\widetilde{\mathbf{M}}}$-inj, and let W denote the class of weak equivalences of $\mathbf{M}$. According to [Hir03, Theorem 11.3.1], we must verify the following (In the case of semi-model categories, we use Proposition A.13):

(a) $J_{\widetilde{\mathbf{M}}}\text{-cof} \subset \big(I_{\widetilde{\mathbf{M}}}\text{-cof}\big) \cap W$. (In the case of semi-model categories, we replace $J_{\widetilde{\mathbf{M}}}$-cof by its subclass of maps with cofibrant domain.)
(b) $I_{\widetilde{\mathbf{M}}}\text{-inj} \subset \big(J_{\widetilde{\mathbf{M}}}\text{-inj}\big) \cap W$.
(c) $\big(J_{\widetilde{\mathbf{M}}}\text{-inj}\big) \cap W \subset I_{\widetilde{\mathbf{M}}}\text{-inj}$.

Before we prove (a) through (c), we observe that the associativity (up to natural isomorphism) of pushout-product implies the following inclusions:

(d) $(I_{\mathbf{V}}\text{-cof})\square\big(I_{\widetilde{\mathbf{M}}}\text{-cof}\big) \subset \big(I_{\widetilde{\mathbf{M}}}\text{-cof}\big)$.
(e) $\big((J_{\mathbf{V}}\text{-cof})\square\big(I_{\widetilde{\mathbf{M}}}\text{-cof}\big)\big) \subset J_{\widetilde{\mathbf{M}}}\text{-cof}$.
(f) $\big((I_{\mathbf{V}}\text{-cof})\square\big(J_{\widetilde{\mathbf{M}}}\text{-cof}\big)\big) \subset J_{\widetilde{\mathbf{M}}}\text{-cof}$.

Note that (d) implies:

(d′) A map p of $\mathbf{M}$ is an $I_{\widetilde{\mathbf{M}}}$-injective if and only if, for each $I_{\widetilde{\mathbf{M}}}$-cofibration i, the map $\langle i, p\rangle$ is a trivial fibration of $\mathbf{V}$.

(There is a similar characterization of $J_{\widetilde{\mathbf{M}}}$-injectives, but we will not need it in the sequal.)

We also observe that:

($*$) The functor $- \otimes \widetilde{\mathbf{1}}\colon \mathbf{M} \to \mathbf{M}$ preserves and detects weak equivalences.
($**$) The morphism $\kappa \otimes \widetilde{\mathbf{1}}$ is a $J_{\mathbf{M}}$-cofibration.

Indeed, assertion ($*$) is clear, because the functor $- \otimes \widetilde{\mathbf{1}}\colon \mathbf{M} \to \mathbf{M}$ is naturally weakly equivalent to the identity functor. Assertion ($**$) is a consequence of the fact that $\kappa \otimes \widetilde{\mathbf{1}}$ is an $I_{\mathbf{M}}$-cofibration (being the composite of a pushout of $\emptyset \to \widetilde{\mathbf{1}}$ and the map $\iota \otimes \widetilde{\mathbf{1}}$) and is a weak equivalence by ($*$).

Now we prove (a) through (c). We start from (a). To show that $J_{\widetilde{\mathbf{M}}}\text{-cof} \subset I_{\widetilde{\mathbf{M}}}\text{-cof}$, it will suffice to show that $J_{\widetilde{\mathbf{M}}} \subset I_{\widetilde{\mathbf{M}}}\text{-cof}$. For this, in light of (d), it suffices to show that κ is an $I_{\widetilde{\mathbf{M}}}$-cofibration. This is clear, because it is the composite of the pushout of ϕ and the $I_{\mathbf{M}}$-cofibration ι. To show that $J_{\widetilde{\mathbf{M}}}\text{-cof} \subset W$, we observe that $\big(J_{\widetilde{\mathbf{M}}}\text{-cof}\big) \otimes \widetilde{\mathbf{1}} \subset J_{\mathbf{M}}\text{-cof}$ by ($**$). It follows from ($*$) that $J_{\widetilde{\mathbf{M}}}\text{-cof} \subset W$, as desired.

Next, we prove (b). Let p be an $I_{\widetilde{\mathbf{M}}}$-injective. We must show that $p \in \big(J_{\widetilde{\mathbf{M}}}\text{-inj}\big) \cap W$. Since $I_{\widetilde{\mathbf{M}}}\text{-inj} \subset I_{\mathbf{M}}\text{-inj} = (J_{\mathbf{M}}\text{-inj}) \cap W$, it suffices to show that $\langle \kappa, p\rangle$ is a trivial fibration and $\langle \phi, p\rangle$ is a fibration. By (d′), this will follow once we show that κ and ϕ are $I_{\widetilde{\mathbf{M}}}$-cofibrations. For ϕ, this is clear. We have also seen in the previous paragraph that κ is an $I_{\widetilde{\mathbf{M}}}$-cofibration. Thus we have proved (b).

Finally, we prove (c). Let $p \in \big(J_{\widetilde{\mathbf{M}}}\text{-inj}\big) \cap W$. We wish to show that $p \in I_{\widetilde{\mathbf{M}}}\text{-inj}$. Since $\big(J_{\widetilde{\mathbf{M}}}\text{-inj}\big) \cap W \subset (J_{\mathbf{M}}\text{-inj}) \cap W = I_{\mathbf{M}}\text{-inj}$, it suffices to show that $\langle \phi, p\rangle$ is a trivial fibration. Since ϕ is a retract of the map $\psi\colon \emptyset \to C$, it suffices to show that $\langle \psi, p\rangle$ is a trivial fibration. For this, factor the map ψ as

$$\emptyset \xrightarrow{\widetilde{\phi}} \widetilde{\mathbf{1}} \xrightarrow{\kappa} C.$$

The map $\widetilde{\phi}$ is an $I_{\mathbf{M}}$-cofibration, so $\left\langle \widetilde{\phi}, p \right\rangle$ is a trivial fibration. The map $\langle \kappa, p \rangle$ is a trivial fibration by the definition of $J_{\widetilde{\mathbf{M}}}$-injectives. Since $\langle \psi, p \rangle$ is the composition of $\langle \kappa, p \rangle$ and a pullback of $\left\langle \widetilde{\phi}, p \right\rangle$, this proves that $\langle \psi, p \rangle$ is a trivial fibration, as desired.

We have thus shown that the sets $I_{\widetilde{\mathbf{M}}}$, $J_{\widetilde{\mathbf{M}}}$ and the class W of weak equivalences of $\mathbf{M}$ gives rise to a model structure on $\mathbf{M}$, which we denote by $\widetilde{\mathbf{M}}$. By construction, it satisfies conditions (1) through (3). We complete the proof by showing that $\widetilde{\mathbf{M}}$ is a monoidal model category; once we prove this, condition (4) is immediate from the construction.

To show that $\widetilde{\mathbf{M}}$ is a monoidal model category, it suffices to show that $I_{\widetilde{\mathbf{M}}} \square I_{\widetilde{\mathbf{M}}} \subset I_{\widetilde{\mathbf{M}}}$-cof, $I_{\widetilde{\mathbf{M}}} \square J_{\widetilde{\mathbf{M}}} \subset J_{\widetilde{\mathbf{M}}}$-cof, and $J_{\widetilde{\mathbf{M}}} \square I_{\widetilde{\mathbf{M}}} \subset J_{\widetilde{\mathbf{M}}}$-cof. Using the fact that $\mathbf{M}$ is a monoidal $\mathbf{V}$-model category, and also using inclusions (a), (d), and (e), we are reduced to showing the following:

$(***)$ For every $I_{\mathbf{M}}$-cofibration i, the maps $i \square \kappa$ and $\kappa \square i$ are $J_{\mathbf{M}}$-cofibrations (and hence $J_{\widetilde{\mathbf{M}}}$-cofibrations). (In the semi-model categorical case, we also assume i has a cofibrant domain in $\mathbf{M}$.)

To prove $(***)$, we observe that $i \square \kappa$ is a composition of $i \square \iota$ and a pushout of $i \otimes \phi \cong i$. Since the maps $i \square \iota$ and i are $I_{\mathbf{M}}$-cofibrations, we deduce that $i \square \kappa$ is an $I_{\mathbf{M}}$-cofibration. Moreover, we know from $(**)$ that $\kappa \otimes \widetilde{\mathbf{1}}$ is an J_M-cofibration, so $(i \square \kappa) \otimes \widetilde{\mathbf{1}}$ is also a $J_{\mathbf{M}}$-cofibration. It follows from $(*)$ that $i \square \kappa$ is a weak equivalence. Hence $i \square \kappa$ is a $J_{\mathbf{M}}$-cofibration. Likewise, $\kappa \square i$ is a $J_{\mathbf{M}}$-cofibration. The proof is now complete. $\square$

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